China Basic Materials

APAC Focus: From Cyclical to Secular: Time to Re-rate HK/China Copper Miners



We believe copper is becoming a secular growth commodity. Electrification, grid investment and AI infrastructure are broadening demand, while declining ore grades, long project lead times and restrained capex constrain supply. Our global team estimates AI data centres and related power infrastructure could contribute 15–20% of global copper demand growth through 2030. Yet HK/China copper miners still trade as cyclical stocks, at roughly half the P/E and a substantial EV/EBITDA discount to overseas peers increasingly valued for structural growth. We see a clear case for HK/China copper miners to re-rate.

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AIDC adds 15–20% of copper demand growth, offsetting lower intensity

AI data centres are not a step change for copper demand, but the incremental demand is sustainable. Our global team estimates demand from data centres and related power generation will rise from 900kt in 2025 to 1.55mt in 2030, contributing 15–20% of global demand growth. While we estimate 800VDC could reduce copper intensity from around 30t/MW to 21–22t/MW during 2026–30E, capacity growth should more than offset the decline. Aluminium substitution risk remains limited. Most suitable grid applications have already switched, while critical grid equipment remains copper-intensive. Adoption in appliances remain selective due to reliability requirements, particularly in China. In autos, substitution is feasible in some power cables and battery busbars, but remains difficult in motors, battery foil and compact wiring.

Supply growth remains constrained

We expect copper supply to remain tight over the next 3–5 years and potentially longer. Ore grades have fallen by roughly 50% over two decades, disruptions are rising and new projects require more capital. Inflation-adjusted capex remains around 30% below the previous peak, while future projects are around 65% more capital-intensive, requiring an estimated US$175bn by 2035E. China's cumulative concentrate imports have turned negative on a YoY basis since April, marking the first contraction since 2021, while depressed treatment charges and slowing refined production signal acute feedstock shortages. China's inventories are at record lows. Although US inventories have risen to over 800kt, we see this as manageable, with data-centre demand alone potentially absorbing 500–600kt.

Beyond US$13,000/t: HK/China miners are not priced for the new norm

Structural demand and constrained supply support US$13,000/t as a new copper price norm rather than a cyclical peak. Yet HK/China copper miners trade average at 8x P/E and 6x EV/EBITDA, roughly half the P/E and well below the average 10x EV/EBITDA of overseas peers. We see meaningful re-rating potential. In our view, Zijin offers strong execution, visible production growth and exposure to gold & lithium; CMOC combines high-grade resources with strong volume delivery; & Jiangxi Copper offers domestic

This report has been prepared by UBS Securities Asia Limited. **ANALYST CERTIFICATION AND REQUIRED DISCLOSURES, INCLUDING INFORMATION ON THE QUANTITATIVE RESEARCH REVIEW PUBLISHED BY UBS, BEGIN ON PAGE 39.**

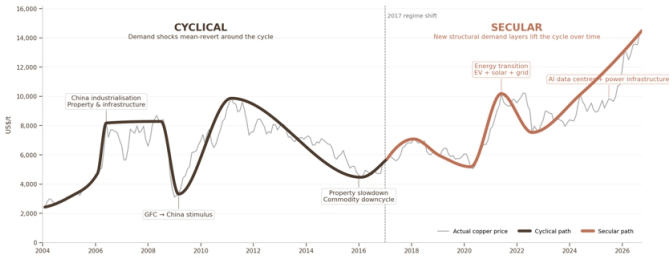
copper exposure with upside from First Quantum, SolGold and its copper foil business.

Figure 1: Summary of our covered copper miners PT and estimate changes

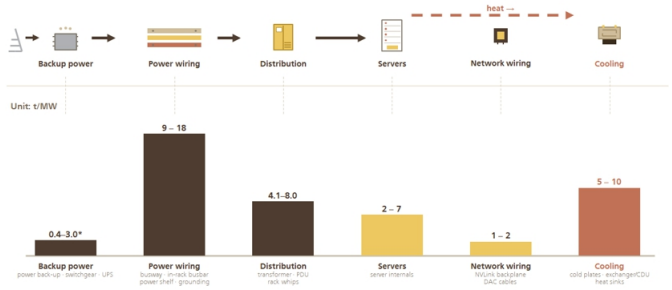
Company	Ticker	Rating	Price Target (LC)				26E earnings (Rmb mn)			27E earnings (Rmb mn)			28E earnings (Rmb mn)			Comment
		New	Upside	New	Old	Chg.	New	Old	Chg.	New	Old	Chg.	New	Old	Chg.	
Zijin-A	601899.SS	Buy	67%	50.00	52.00	-4%	81,414	82,912	-2%	93,605	89,208	5%	102,263	96,342	6%	Slightly Lower 2026 estimates based on gold price adjustment down and cost inflation. SOTP based on 15x 2027E target P/E for copper and 20x 2027E target P/E for gold.
Zijin-H	2899.HK	Buy	75%	56.00	57.80	-3%	81,414	82,912	-2%	93,605	89,208	5%	102,263	96,342	6%	
CMOC-A	603993.SS	Buy	74%	29.00	31.00	-6%	32,186	32,048	0%	34,993	33,618	4%	38,943	36,011	8%	Lower PT based on country risk discount of 10%. PE based on 20x 2027E target P/E.
CMOC-H	3993.HK	Buy	110%	32.00	33.00	-3%	32,186	32,048	0%	34,993	33,618	4%	38,943	36,011	8%	
JCC	0358.HK	Buy	100%	65.00	65.00	0%	16,081	14,968	7%	17,158	15,875	8%	18,282	17,207	6%	Slightly lift 2026 estimates based on sulphuric acid price increase. PE based on 12x 2027E target P/E.

Source: UBS estimates

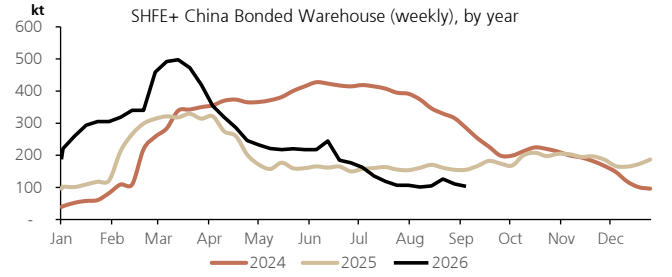
OUR THESIS IN PICTURES



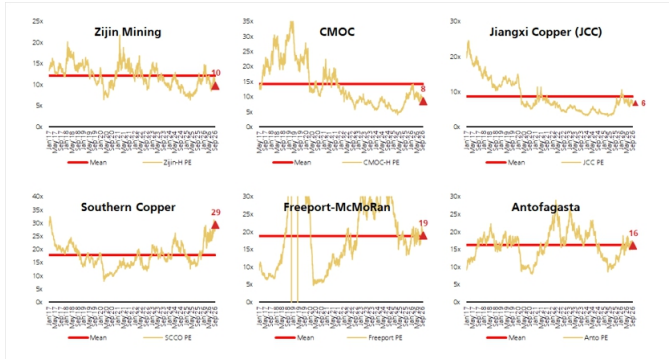
Copper is shifting from a cyclical peak to a higher structural price range.



AI data centres add a durable source of copper demand.



Record-low China inventories suggest tighter supply constraints.



Structural copper, cyclical valuations: HK/China miners trade at half the P/E of overseas peers.

Sources for exhibits above: Company data, UBS Research

UBS Research THESIS MAP MOST FAVORED LEAST FAVORED

CMOC, Zijin, Jiangxi Copper

PIVOTAL QUESTIONS

Q: Will AIDC and power infrastructure support copper demand?

Yes, we think so. Our global team estimates copper demand from data centres and associated power generation will rise from 900kt in 2025 to 1.55mt in 2030, contributing 15–20% of global copper demand growth. Although 800VDC could reduce copper intensity per MW, growth in new capacity should more than offset this decline, making AIDC a durable source of incremental demand.

Q: Will copper supply remain tight over the next five years?

Yes. Declining ore grades, long project lead times, execution risks and years of restrained capex should limit major miners to moderate production growth. Depressed treatment charges, slowing Chinese smelter output and low global inventories suggest the market is already tight, with limited buffer against disruptions or stronger-than-expected demand.

Q: Should HK/China copper miners re-rate?

Yes. We believe US$13,000/t could become the new copper price norm as structural demand growth meets a slow supply response. Yet HK/China copper miners remain valued largely as traditional cyclicals, unlike overseas peers increasingly priced for copper's structural growth, creating re-rating potential for Zijin, CMOC and Jiangxi Copper..

WHAT'S PRICED IN?

Current valuations still price HK/China copper miners as traditional cyclicals rather than beneficiaries of a structural copper market. Despite copper entering a US$13,000/t new norm and our forecasts pointing to historically strong near-term profitability, market-implied returns suggest profitability will fall back towards the average of the past five years. This contrasts with overseas peers, whose valuations increasingly reflect tighter long-term supply and structural demand growth.

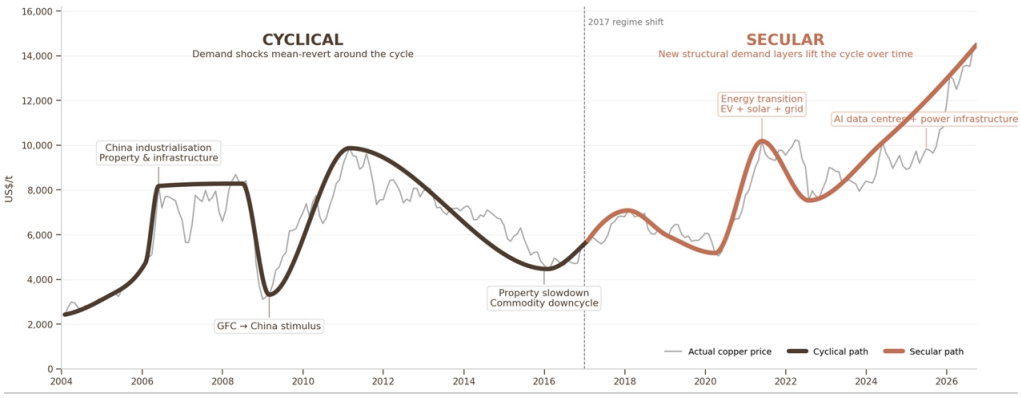
UBSVIEW

The market underestimates how long copper fundamentals can stay strong and continues to value HK/China miners through a cyclical lens. AI and power infrastructure add durable demand on top of electrification and grid investment, while declining ore grades, long project lead times and restrained capex limit the supply response. We therefore see US$13,000/t as a new norm rather than a cyclical peak. If copper prices and returns remain higher for longer, we believe the market's expectation of a rapid earnings normalisation will prove too conservative, supporting a potential re-rating of Zijin, CMOC and Jiangxi Copper.

EVIDENCE

The data support stronger-for-longer copper fundamentals, but HK/China miners are not priced accordingly. AIDC and related power demand could contribute 15–20% of global copper demand growth through 2030, while declining ore grades, restrained capex and low inventories constrain supply. Market-implied returns nevertheless revert towards historical averages, giving limited credit to a higher copper price, production growth or company-specific catalysts.

Figure 2: Secular copper, cyclical valuations for HK/China copper miners



Source: UBS estimates

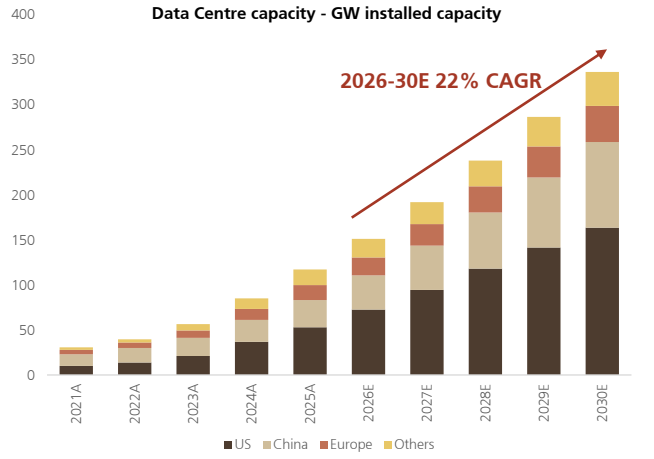
AI data centres: a new source of structural copper demand

Copper demand has historically been shaped by successive investment cycles, from housing and construction to electrification and the energy transition. We see **AI infrastructure as the next source of structural demand growth**.

Data centres and their associated power infrastructure are among the most direct beneficiaries of AI-related capital spending. They require substantial investment in power generation, grid connections, transformers, distribution equipment and cooling systems, all of which are copper-intensive. This brings copper firmly into the AI investment theme. We do not expect AI data centres alone to transform the global copper market. Instead, they add another durable source of demand alongside electric vehicles, renewable energy and grid investment. Copper typically represents a small share of total project cost in these applications, while substitution can compromise efficiency, reliability or space utilisation. We therefore expect demand elasticity to remain limited over the next three to five years.

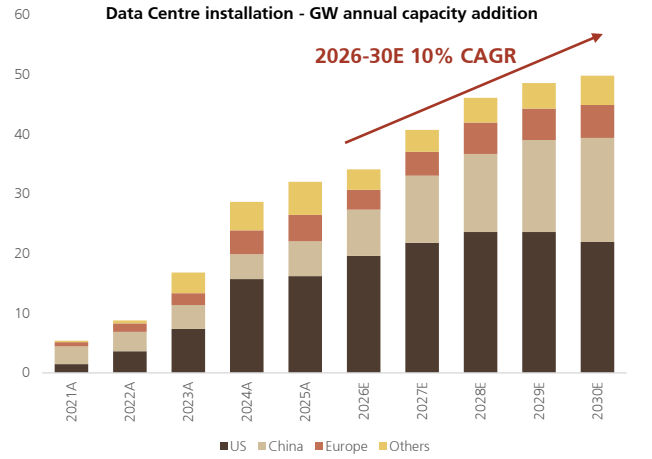
Our global team estimates that data centres and associated power generation accounted for c.900kt of copper demand in 2025, comprising around 700kt from data centres and 200kt from incremental power generation. We estimate demand will rise to approximately 1.55mt by 2030, implying an 11% CAGR. This would increase the sector's share of global copper demand from around 3% to 4% and contribute 15–20% of global copper demand growth over the period. In the US, where AI infrastructure investment is particularly concentrated, data centres and related power infrastructure could account for roughly one-quarter of copper demand growth.

Figure 3: Global team forecast of data Centre installation - GW installed capacity



Source: UBS Research, UBS Evidence Lab

Figure 4: Global team forecast of data Centre installation - GW annual capacity addition



Source: UBS Research, UBS Evidence Lab

Copper intensity matters more than copper content per rack

AIDC copper demand depends on two factors: how much new capacity is commissioned each year and how much copper is required per MW of that capacity. The key question is therefore not simply how many AI racks are shipped or how much copper each rack contains, but whether capacity growth can outpace the decline in copper intensity. This distinction matters because two apparently conflicting trends can occur at the same time. As rack power density rises, liquid cooling becomes more prevalent and board-level power distribution becomes more complex, the amount of copper in each rack may increase. However, higher-power racks also mean that far fewer racks are required to support the same 1MW of IT load. Meanwhile, 800VDC power architecture, aluminium conductors and optical interconnects can further reduce copper use per unit of installed capacity.

The industry could therefore see more copper per rack but less copper per MW. The two trends are not contradictory because they use different denominators. UBS Capital Goods team expects installed data-centre capacity to grow at a 22% CAGR during 2026–30E, while annual capacity additions rise at around a 10% CAGR, supported by UBS evidence lab data. We estimate facility-level copper intensity could decline from approximately 30t/MW today to 21–22t/MW by 2030, but capacity growth should more than offset the decline. As a result, absolute annual copper demand from data centres should continue to increase.

Where is copper used in a data centre?

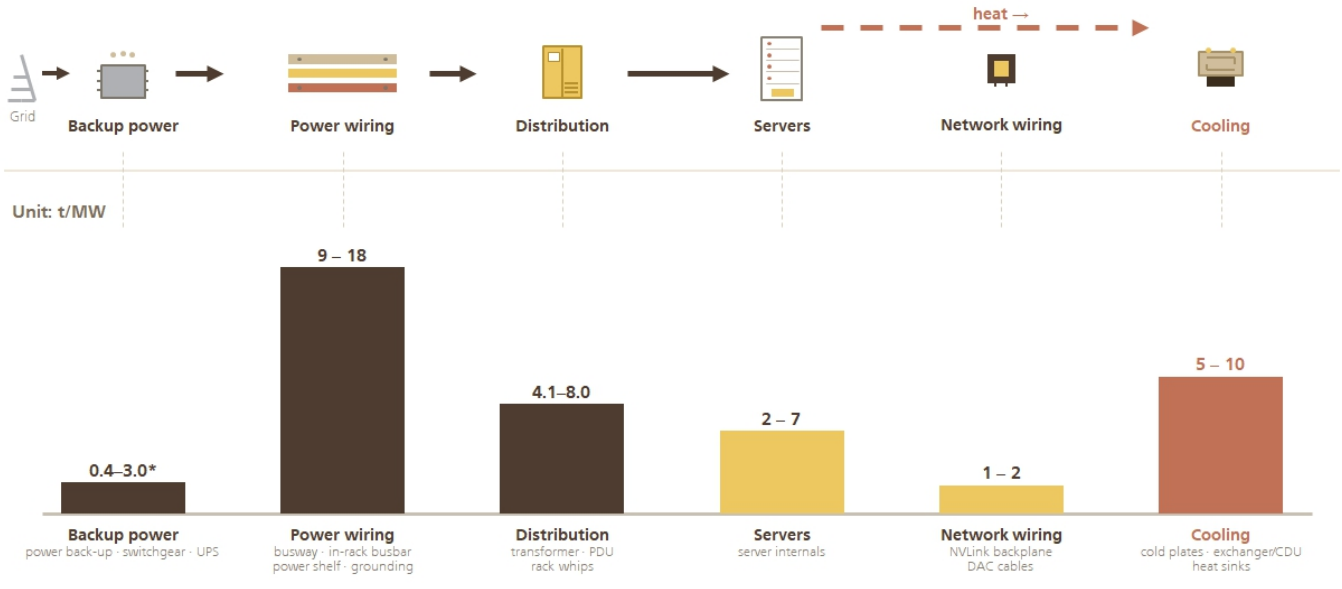
To understand how technology changes affect copper intensity, we divide facility-level copper demand into three main categories:

Copper in a data centre follows three functions: power, cooling and data transmission.

- Power infrastructure (13.5-29 t/MW, c.63% of total) includes transformers, switchgear, UPS systems, backup power equipment, busways, feeder cables, PDUs and rack-level power distribution.
- Cooling infrastructure (5-10 t/MW, c.22%) includes chillers, pumps, heat exchangers, cooling distribution units (CDUs) and liquid cooling equipment.
- Interconnects (3-9 t/MW, c.15%) include network cabling within and between racks, halls and campus facilities.

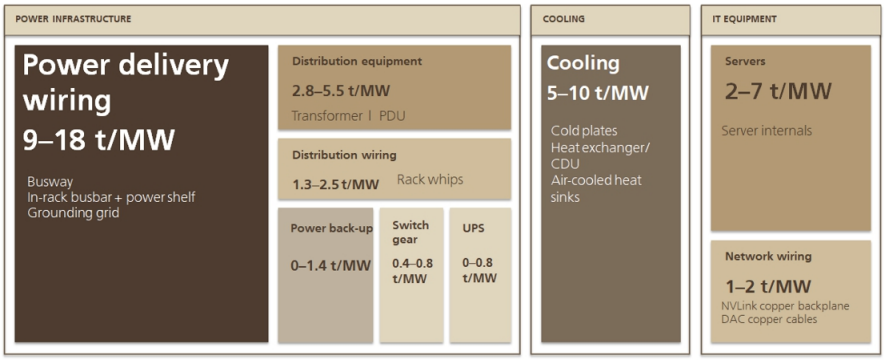
Power infrastructure is by far the largest copper pool, accounting for roughly two-thirds of total copper use. Cooling is the second-largest component, and networking and interconnects remain the rest. Different technology trends affect different copper pools. 800VDC primarily reduces power-distribution copper, liquid cooling increases cooling-related copper, aluminium substitution targets large-section conductors, while optical networking only affects a relatively small potion of interconnect copper. As a result, changes in copper intensity are driven mainly by what happens to the power infrastructure layer rather than the networking layer.

Figure 5: Copper flows through AI data-centre power, cooling and interconnect infrastructure



Source: S&P Global, UBS estimates

Figure 6: Power infrastructure accounts for c.63% of data-centre copper intensity



Source: S&P Global, UBS estimates

Higher rack density lowers the number of racks per MW

Rack power density is increasing rapidly as AI computing platforms become more powerful. A representative HGX deployment operates at around 45kW per rack, while GB200 increases rack power to approximately 120kW. VR200 is expected to move beyond 200kW, with future VR300 with 800VDC/Feynman systems moving towards several hundred kilowatts and potentially MW-scale configurations.

Our bottom-up estimates indicate that rack-level copper content rises with power density:

- c. 30–80kg for a HGX rack
- c. 120–275kg for GB200
- c. 175–385kg for VR300
- More than 300kg for future 800 VDC configurations

However, the number of racks required per MW falls sharply. Supporting 1MW of IT load requires around 22 racks at 45kW, 8.3 racks at 120kW, but only 1.7 racks at 600kW. A fifteen-fold increase in rack power therefore reduces the number of racks per MW by more than 90%. This does not translate into a comparable decline in overall copper intensity. Much of the copper in a data centre is linked to installed power capacity rather than rack count. Transformers, switchgear, UPS systems, facility-level cables and cooling equipment are largely sized according to MW of load. Higher rack density reduces rack-dependent copper, but does not eliminate the majority of facility-level copper.

We therefore view higher rack density as an enabling factor for lower copper intensity, rather than the main driver. The larger reduction comes when higher-density racks are combined with 800VDC architecture, centralised power distribution and greater infrastructure sharing. This explains why rack numbers can fall by more than 90%, while total facility-level copper intensity declines by a more moderate 25–30%, from around 30t/MW to 21–22t/MW.

Figure 7: Higher rack power raises copper per rack but lowers copper per kW

NIVIDIA Model	Rack power consumption (kW)	Copper Inside The Rack (kg)	mid point	Copper kg/kW	Launch / Deployment
HGX	45	**29-80**	55	1.2	Released in 2022 / deployed from 2023
GB200	120	**120~275**	198	1.6	First deliveries in Q4 2024 / scaled deployment in 2025
GB300	140	**129~292**	210	1.5	H2 2025
VR200	227	**175~385**	280	1.2	H2 2026
VR300 - 800 VDC not ready	450	**245~503**	374	0.8	H2 2027
VR300 - 800 VDC ready	450	**201~441**	321	0.7	H2 2027

Source: Company data, UBS estimates

Figure 8: AI rack architecture is shifting towards higher power density and 800VDC



Source: company data, UBS estimates

Other factors largely offset, leaving 800VDC as the key driver

Our intensity bridge helps assess the relative importance of the different factors affecting AIDC copper intensity. Liquid cooling, higher redundancy and regional mix increase copper use, while higher rack density, aluminium substitution and optical migration reduce it. Excluding 800VDC, these factors broadly offset one another, leaving facility-level copper intensity close to today's level of around 30t/MW.

Liquid cooling is one of the most important upside factors. As rack power density rises, greater liquid-cooling penetration adds copper through cold plates, pumps, motors, heat exchangers and cooling distribution units. However, it can also reduce the need for server fans, air-handling equipment, mechanical cooling capacity and white space per MW. On balance, we estimate liquid cooling increases facility-level copper intensity by approximately 0–2t/MW, with a base-case impact of around 1t/MW. Cold-plate cooling should remain the dominant architecture in the near term. Based on our channel checks with **Hailiang** and discussions with the **UBS Chemicals team**, we view immersion cooling as a highly effective thermal-management solution with significant long-term potential. However, near-term commercial adoption is likely to remain limited by infrastructure requirements, ecosystem readiness and deployment complexity. Higher redundancy requirements and regional mix provide further upside by increasing duplicated power and cooling infrastructure.

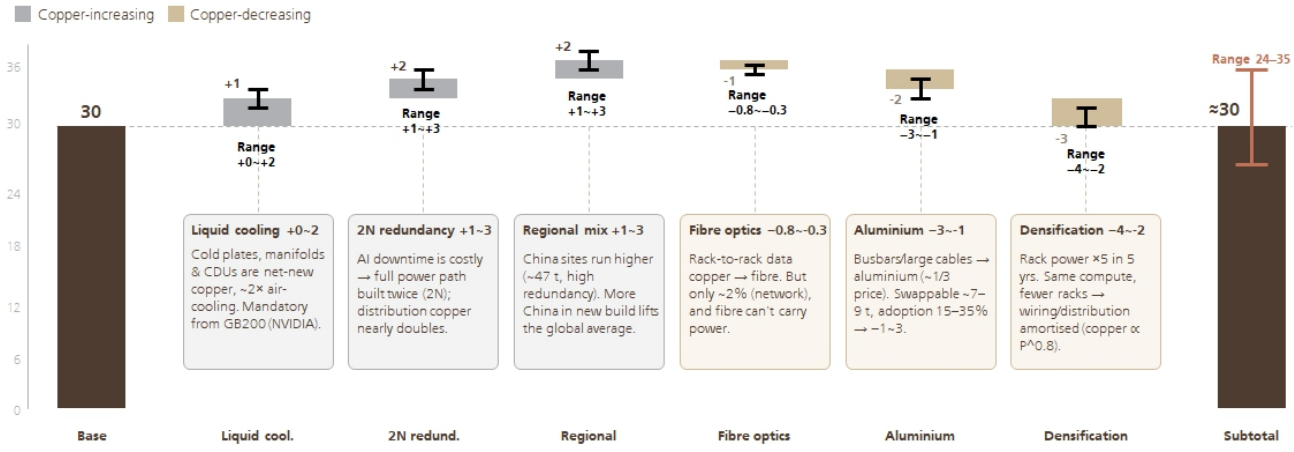
The downside factors are more dispersed. Higher rack density reduces rack-dependent infrastructure per MW, aluminium replaces copper in selected large-section conductors, and optical migration reduces copper in data transmission. However, aluminium and optical technologies affect only parts of the overall copper pool, limiting their impact on total facility-level intensity.

Figure 9: At a Glance: How AIDC Drivers Affect Copper Use

Driver	What physically changes	Copper component	Direction & Range
800VDC	Supply voltage rises from 415V to 800V. For the same power, current roughly halves, so conductors can be thinner.	**Main power cables / busbars**	↓ Significant
Liquid cooling	When air cooling is no longer enough, liquid removes the heat. Copper heat sinks give way to cold plates, liquid loops and a CDU.	**Cold plates + liquid loops / CDU**	↑ +0~+2 Modest increase; immersion may replace it after 2028
2N redundancy	To avoid outages, a second power path duplicates part of the distribution system.	**Power equipment + cabling (duplicated)**	↑ +1~+3 More copper from dual power paths
Regional mix	Redundancy standards and construction practices vary by region, shifting the average; more backup capacity means more copper.	**Power distribution (high redundancy)**	↑ +1~+3 More high-redundancy sites
Fibre optics	Fibre replaces copper data cables, but it cannot replace copper used for power.	**Inter-rack copper data cables**	↓ −0.8~−0.3 Only data cabling is affected
Aluminium swap	Large conductors switch from copper to aluminium. Aluminium needs a larger cross-section, so the swap works mainly on major conductors.	**Large busbars and high-current cables**	↓ −3~−1 Some copper conductors switch to Al
Densification	The same compute fits into fewer, more powerful racks (120→600 kW each), reducing inter-rack cabling and pipework.	**Power-distribution cabling between racks**	↓ −4~−2 Wiring and distribution spread over more MW

Source: UBS estimates

Figure 10: Intensity drivers except 800 VDC broadly offset, leaving 800VDC as the key variable



Source: UBS estimates

800VDC is the key driver of lower copper intensity

Among the factors affecting AIDC copper intensity, we see the transition to 800VDC power architecture as the most important. At a given power level, higher voltage reduces the current carried through the distribution system, allowing smaller cables, conductors and busbars and enabling more centralised and efficient power delivery. The impact is particularly significant because power infrastructure accounts for around 70% of total facility-level copper intensity.

Traditional 54V rack-level power distribution becomes increasingly constrained as rack power rises beyond approximately 200kW. At higher power levels, maintaining a low-voltage architecture would require increasingly large conductors and busbars, creating challenges around space, heat and efficiency. The transition to 800VDC addresses these

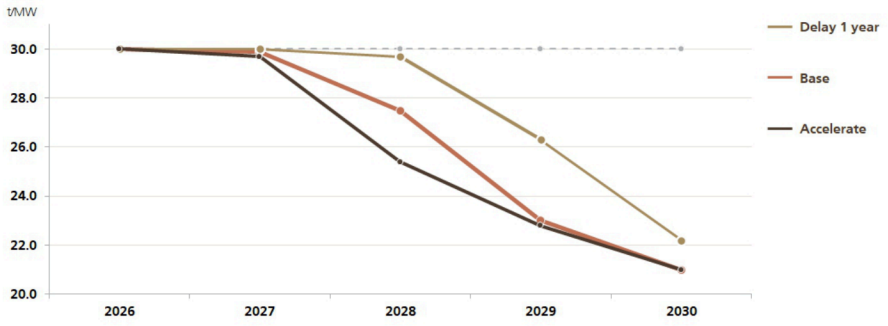
constraints by reducing current requirements and removing part of the low-voltage conversion and distribution infrastructure.

Rather than treating copper intensity as a fixed assumption, we model the transition from the bottom up. Our framework converts annual shipments of major AI platforms, including GB200, GB300, VR200 and future VR300 systems, into newly commissioned capacity using rack-power and PUE assumptions. We then estimate the proportion of new capacity deployed on 800VDC-ready architectures and assess the resulting impact across power distribution, cooling and networking.

VR200/300 or Feynman are not, by themselves, the reason copper intensity declines. Instead, they represent a broader shift in which higher rack power density pushes data centres towards 800VDC architecture. Platform mix determines the pace of this transition, while 800VDC determines the extent of the reduction in power-distribution copper. Based on our deployment assumptions, facility-level copper intensity declines by approximately 25–30% by 2030, **from around 30t/MW today to 21–22t/MW.**

However, lower copper intensity does not imply lower absolute copper demand. Capacity growth remains the larger driver, and the conversion of announced AI investment into operating capacity will depend not only on GPU shipments, but also on power availability, grid connections, transformer supply, generation equipment and project commissioning. In our view, capacity growth should more than offset the decline in copper intensity.

Figure 11: 800VDC implementation timing determines the copper-intensity trajectory



Source: UBS estimates

Optical migration matters for networking, but not for overall copper demand

"Optical in, copper out" is directionally correct for parts of the data-transmission layer, but its effect on total data-centre copper demand is often overstated.

Optical technology does not replace the copper used in transformers, switchgear, UPS systems, busways, power cables, motors or cooling equipment. It acts primarily on networking and interconnects, which represent around 5% of facility-level copper intensity.

Within the data layer, copper use can be divided into three categories:

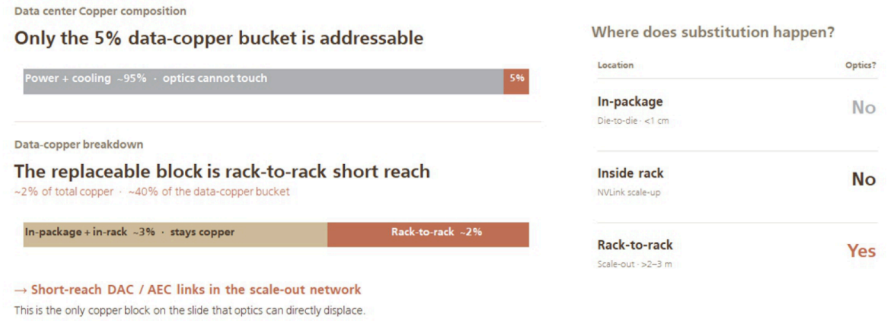
- In-package and board-level interconnects
- In-rack scale-up interconnects
- Rack-to-rack and data-hall scale-out networking

Optical substitution is most advanced in scale-out networking, where fibre increasingly replaces copper over longer distances. Its penetration is more limited in scale-up architectures, where short distances, low latency, power efficiency and cost continue to favour copper.

This distinction is increasingly important. NVIDIA's GB200 and Rubin platforms continue to use copper-based NVLink connections within the rack. As scale-up systems become a larger part of AI infrastructure, data-transmission copper can shift from rack-to-rack networking towards in-rack interconnects. The industry may therefore see less copper in traditional networking, but more copper within increasingly dense AI racks.

We estimate optical migration could reduce facility-level copper intensity by around 0.5t/MW, with a range of approximately 0.3–0.8t/MW. This is equivalent to **roughly 1–3%** of total facility-level copper intensity. Optical migration is highly relevant to networking and optical-component supply chains, but remains a second-order variable for overall data-centre copper demand.

Figure 12: Fiber targets data copper—but the replaceable slice is only ~2% of total copper



Source: UBS estimates

Aluminium substitution: a larger copper pool, but a smaller driver than 800VDC

Unlike optical migration, which affects data-transmission copper, aluminium substitution targets power-distribution copper, the largest copper pool within a data centre. However, aluminium substitution primarily changes the material used rather than reducing the underlying conductor requirement.

The highest substitution potential lies in large-section conductors such as busbars, busways and feeder cables, where space constraints are relatively limited and aluminium technology is already well established. By contrast, copper remains the preferred material in transformers, UPS magnetics, rack-level power distribution and other high-density applications where efficiency, thermal performance, connection reliability and space utilisation are critical. As rack power density increases, these constraints become more important, making aluminium less suitable for the most demanding parts of the power chain.

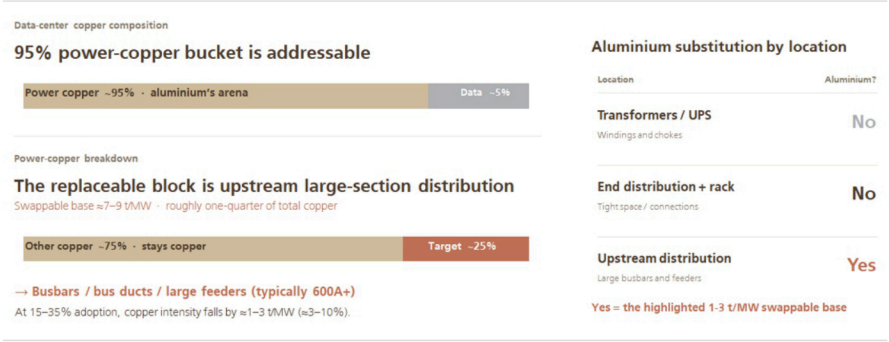
The economic case for aluminium is straightforward. Aluminium conductors are commercially available at scale and typically cost a fraction of copper on a volume basis. However, aluminium does not eliminate conductor requirements. Because its conductivity is lower than copper, an equivalent conductor requires a larger cross-sectional area. As a result, aluminium substitution reduces copper demand but does not reduce the amount of power infrastructure required.

We estimate aluminium substitution could reduce facility-level copper intensity by approximately 1–3 t/MW, equivalent to roughly 3–10% of total copper intensity. This is a meaningful reduction, but still materially smaller than the impact of 800VDC adoption.

Importantly, aluminium substitution and 800VDC act on many of the same conductors and therefore should not be added together mechanically. Conceptually, the effects are sequential rather than additive: 800VDC determines the conductor requirement at a given power load, while aluminium substitution determines how much of the resulting conductor base remains copper. In practice, both decisions may be incorporated into the same system design.

In our view, aluminium substitution is best understood as a second-stage optimisation. Power-architecture changes such as 800VDC determine how much conductor is required; aluminium substitution determines how much of the remaining conductor continues to use copper. As a result, while aluminium substitution is directionally important, it is unlikely to be the primary driver of future changes in AI data centre copper intensity.

Figure 13: Aluminum replacing power copper, can replace c.3–10% of facility copper, less than the impact of 800VDC



Source: UBS estimates

Capacity growth should outweigh falling copper intensity

The central conclusion is not that AIDC copper intensity will remain unchanged. We expect it to decline as higher rack density, 800VDC architecture, optical migration and aluminium substitution improve infrastructure efficiency. However, copper intensity is only one side of the equation. Our global team expects a large and sustained increase in newly commissioned data-centre capacity, supported by continued demand for AI compute and investment in associated power infrastructure. Even with a 25–30% decline in copper intensity, from approximately 30t/MW to 21–22t/MW, capacity additions should be sufficient to lift annual copper demand from 900kt in 2025 to 1.55Mt in 2030.

We therefore see AIDC as a durable source of structural copper demand, but not because copper content per rack or copper intensity remains constant. The demand case rests on capacity growth being strong enough to more than offset improving copper efficiency.

Aluminium substitution depends on total system cost, not metal prices alone

Beyond AI data centres, we assess the substitution risk across power grids, home appliances and autos. Our conclusion is that aluminium will continue to gain share in price-sensitive and less technically demanding applications, but broad-based copper replacement remains unlikely. The feasible substitution rate depends not only on relative metal prices, but also on conductivity, space, weight, thermal performance, mechanical strength, connection reliability and switching costs.

The economic incentive to substitute is clearly rising. Industry participants typically begin to evaluate aluminium when the copper-to-aluminium price ratio reaches around 3.5–4.0. The ratio currently reaches a record 4.9, while aluminium costs roughly one-quarter as much as copper by weight. These conditions make aluminium increasingly attractive where material cost is the primary consideration.

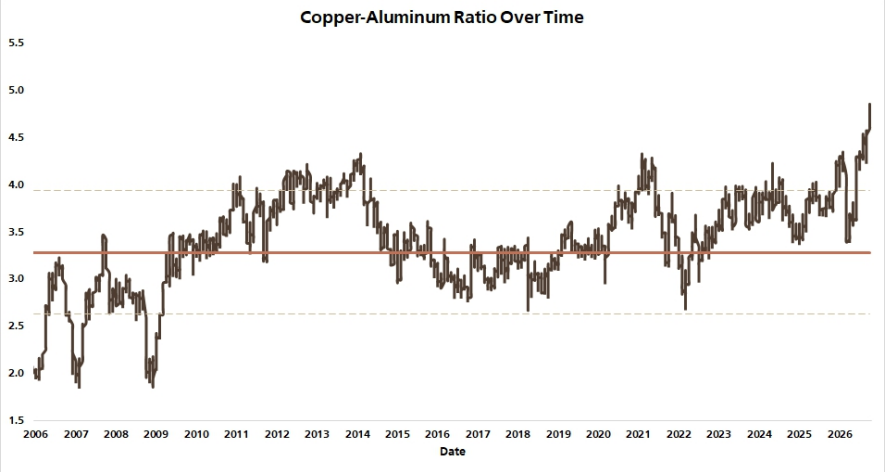
However, the comparison by weight overstates the potential saving. Aluminium has only around 61% of copper's electrical conductivity, meaning an aluminium conductor requires approximately 1.6 times the cross-sectional area to carry the same current. The resulting conductor is larger and requires more installation space, while processing, joining and terminal design can be more complex. Aluminium also differs from copper in thermal conductivity, mechanical strength, thermal expansion and connection stability,

with potential implications for energy losses, heat dissipation, durability and maintenance.

The relevant comparison is therefore total installed and lifetime system cost, rather than metal cost alone. Aluminium is most competitive where conductor size is less constrained, weight reduction is valuable and technical requirements are moderate. Copper remains harder to replace where space, efficiency, heat management, corrosion resistance and long-term connection reliability are critical. A higher copper-to-aluminium price ratio should accelerate selective substitution, but it does not imply uniform adoption across applications.

This framework explains why substitution differs significantly across downstream markets. In power grids, most readily substitutable overhead conductors already use aluminium; in home appliances, adoption is concentrated in selected heat exchangers and export products; and in autos, the clearest opportunities are in larger power cables and battery busbars. Applications such as underground and offshore cables, large transformers, compressors, motor windings, battery copper foil and compact wiring systems should remain more copper-intensive.

Figure 14: Copper-aluminum ratio



Source: Wind, UBS estimates, as of Oct 2026

Figure 15: Aluminium's lower price is offset by lower conductivity and larger conductor requirements

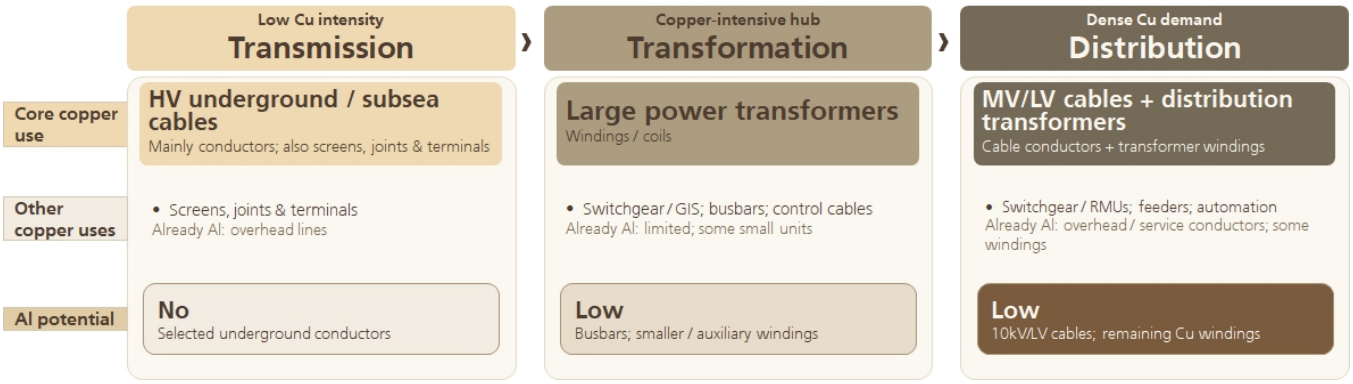
Performance Dimension	Copper	Aluminum
Electrical Conductivity & Transmission	Copper conductivity is approximately 1.64× that of aluminum.	• To carry the same current, an aluminum conductor generally needs a cross-sectional area 30–50% larger than copper. • Aluminum conductors generate more heat, implying higher electrical energy losses.
Mechanical Strength & Workability	Copper tensile strength is approximately 2.5–3.0× that of aluminum.	• Softer material; repeated bending can cause cracking because of weaker fatigue resistance. • Greater thermal expansion and contraction. Repeated heating during operation and cooling during shutdown can cause creep (slow deformation), which may loosen connections. • Density is only about 30% of copper (roughly one-third), giving aluminum a major weight-saving advantage in aircraft, vehicles, and high-voltage transmission towers.
Chemical Stability & Safety	Copper oxide remains conductive. Copper also has good corrosion resistance, so connections are generally stable.	• Aluminum oxidizes readily, forming a hard, electrically insulating oxide film with very high resistance. This can impede current flow and overheat joints. • Direct copper-to-aluminum connections can suffer severe galvanic corrosion in humid environments because of the large electrochemical potential difference, leading to rapid connection failure.

Source: UBS estimates

Power grids: substitution risk is limited, while the capex mix remains supportive

We see limited downside to grid copper demand from further aluminium substitution. This is because most technically straightforward substitution has already occurred, particularly in overhead transmission lines, including UHV lines, where aluminium and aluminium-alloy conductors are already widely used. The remaining substitution opportunity is concentrated in lower-voltage distribution cables and selected smaller, cost-sensitive transformers.The scope for substitution differs across the three layers of the grid. Transmission has the highest existing aluminium penetration and therefore limited incremental substitution potential. Within transformation equipment, aluminium can be used in selected busbars and smaller transformers, but large power transformers, converter stations and other critical equipment remain copper-intensive. Distribution has the largest remaining substitution pool, mainly in low-voltage cables and selected 10kV applications. However, medium- and high-voltage cables, underground and offshore cables, and other space- or reliability-sensitive applications should remain predominantly copper-based because of stricter requirements for conductivity, efficiency, corrosion resistance and long-term reliability.

Figure 16: Copper use in power grid: transmission,transformation, distribution



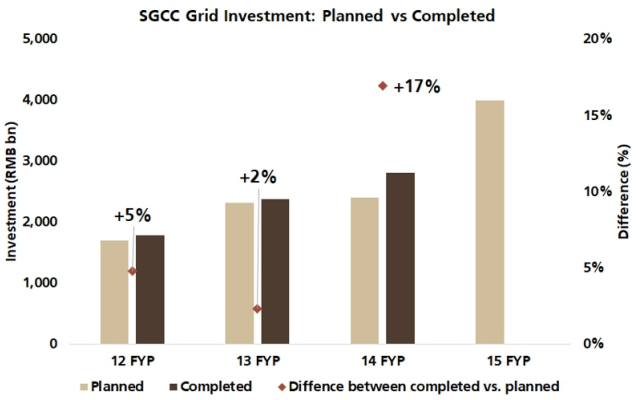
Source: UBS estimates

National level: The capex mix matters more than the headline number

China's grid investment outlook remains supportive, with national grid fixed-asset investment expected to exceed RMB5tn during 2026–30E. However, the impact on copper demand will depend more on where this capital is deployed than on the headline investment figure.

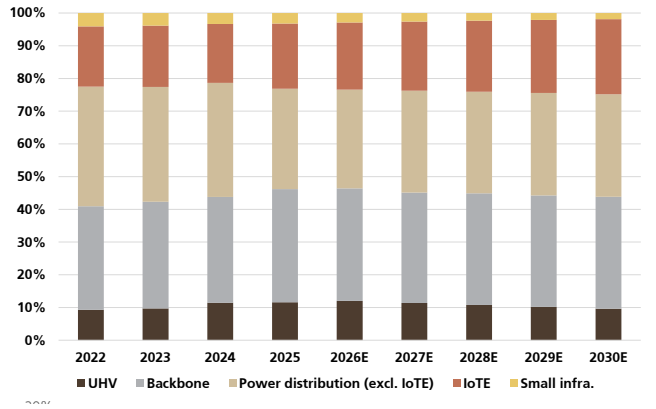
Based on UBS Utilities estimates, around 10% of investment could be allocated to UHV and 30% to backbone networks, close to 40% to power distribution excluding IoT, around 15% to grid digitalisation and IoT, with the remainder allocated to smaller infrastructure. These categories have very different copper intensities. Overhead transmission corridors are relatively aluminium-intensive, while underground distribution networks, substations, converter stations and transformer capacity are more copper-intensive. By contrast, spending on land, civil works and digital systems has relatively limited direct metal content.This means that grid capex growth does not translate mechanically into copper demand growth. However, the sizeable allocation to distribution upgrades and transformation equipment remains supportive, as these areas contain a higher share of underground cables, transformers, substations and other copper-intensive infrastructure.

Figure 17: SGCC grid investment is expected to exceed Rmb4tn in 2026–30E, and UBS expects 2026-30E CAGR 11%



Source: CEC, State Grid, China Southern Grid, UBS estimates

Figure 18: Grid capex mix matters more than headline investment for copper demand



Source: CEC, State Grid, China Southern Grid, UBS estimates

Industry and company evidence point to selective substitution

Industry feedback suggests that incremental aluminium substitution will remain concentrated in low-voltage and less technically demanding applications. **SMM** expects substitution to slow as penetration rises and engineering constraints become more binding. This implies an incremental substitution-risk ranking of distribution first, followed by transformation and then transmission. Transmission already has the highest aluminium penetration, while distribution has the largest remaining addressable pool. Company evidence supports this measured pace of substitution. **Prysmian's** aluminium share has increased by around 3 percentage points over the past five years to approximately 40% by weight, while copper still represents around 60%. **Nexans** expects grid investment to support demand for both metals, with copper remaining the benchmark for technically demanding applications and aluminium gaining share where cost, weight and availability matter more.

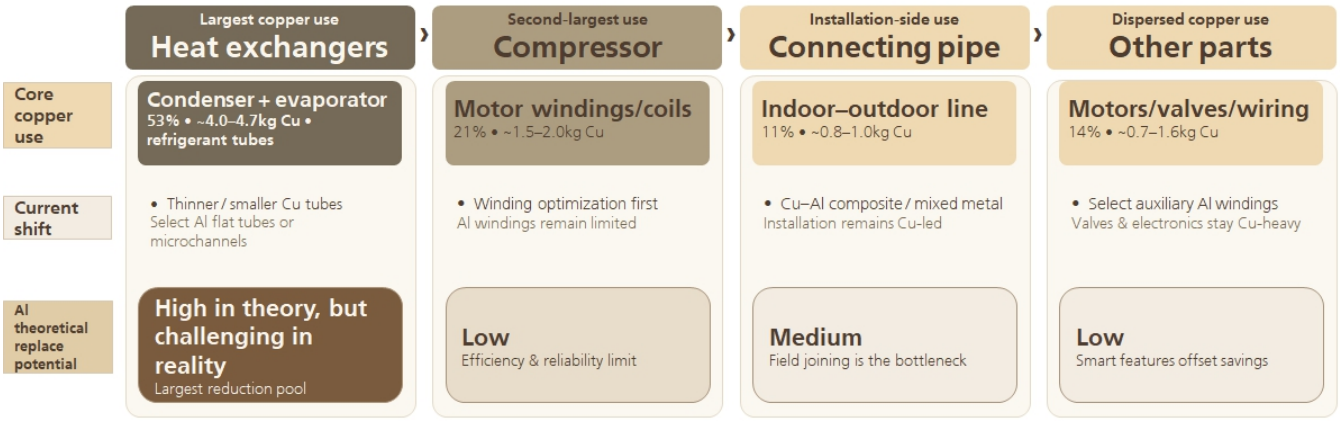
Our channel checks also suggest that **Dongfang's** offshore cables remain almost entirely copper-based, while land-based cables offer greater flexibility for aluminium adoption. Offshore systems face stricter requirements for corrosion resistance, reliability and maintenance, making copper more difficult to replace. In addition, most cable manufacturers hedge their metal costs, reducing the need to switch conductor materials solely in response to short-term copper-price movements.

Overall, we expect aluminium substitution to reduce grid copper intensity only at the margin. Most readily substitutable transmission applications have already moved to aluminium, while the remaining opportunity is concentrated in low-voltage distribution. Continued investment in underground cables, substations, converter stations and large transformers should therefore remain supportive of grid copper demand.

Home appliances: Aluminum substitution is commercially proven in selected applications, but progress remains highly component- and region-specific, and cautious

According to UBS consumer analyst Rennie Pan and industry data, a typical 1.5HP residential air conditioner contains about 8kg of copper. Heat exchangers account for the largest share, including approximately 35% in the condenser and 18% in the evaporator. Compressors account for around 21%, connecting pipes for 11%, and other components, including motors, valves, wiring and electronics, for 14%.

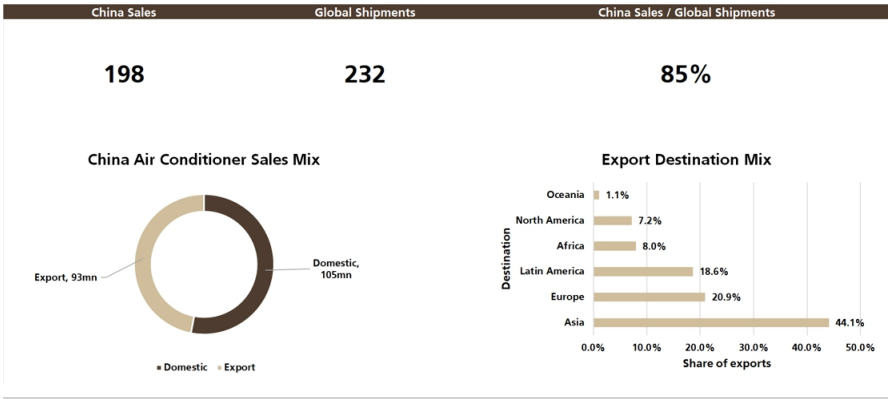
Figure 19: Copper usage in air conditioner



Source: UBS estimates

China is central to global air-conditioner production and trade. In 2025, China sold 198 million units, including 105 million units domestically and 93 million units overseas. Exports accounted for 47% of Chinese sales, with Asia representing 44% of export volume, Europe 21%, Latin America 19%, Africa 8%, North America 7% and Oceania 1%. China accounted for approximately 75% of global sales and 70% of global production.

Figure 20: Home appliances overview: shipments, export, import



Source: Industry data, UBS estimates

Industry expert view

Copper intensity and aluminum adoption vary by region. Progress in China's domestic market has been limited by consumer concerns and manufacturers' focus on product reliability and after-sales service. Adoption appears to be more advanced in export markets, where aluminum components face less consumer resistance and provide greater cost benefits.

Japan provides evidence that aluminum substitution is technically and commercially feasible, but not yet universal. The most mature applications are in outdoor-unit heat exchangers and selected refrigerant pipes. Compressors, motor windings, indoor-unit evaporators and critical joints remain more dependent on copper. Japanese manufacturers have commercialized all-aluminum or aluminum flat-tube heat exchangers in selected commercial and outdoor-unit products. The industry is moving from conventional copper-tube and aluminum-fin structures toward all-aluminum microchannels and mixed-material designs using copper, aluminum and stainless steel according to function. However, disclosure of residential penetration, model coverage and actual copper savings remains limited.

Some industry sources estimate that all-aluminum heat exchangers have reached 40% to 50% penetration in Japanese residential air conditioners. The scope and denominator of these estimates remain unclear. **Our channel check with the Daikin coverage**

analyst also suggests that Daikin remains cautious, as differences in machinability, conductivity and reliability continue to limit direct replacement of copper.

Company-level progress

Daikin. Daikin has developed an all-aluminum microchannel heat exchanger capable of both cooling and heating. The technology uses flat aluminum tubes and integrated refrigerant channels to improve heat transfer, reduce weight and size, and lower refrigerant requirements. Key development challenges included drainage during heating, the distribution of two-phase refrigerant and corrosion resistance.

Daikin's recent product design also indicates a selective rather than complete substitution strategy. In some commercial outdoor units, aluminum connecting pipes and larger stainless-steel pipes are used alongside small-diameter internally grooved copper tubes. Copper remains in locations requiring high thermal performance and precise processing, while aluminum and stainless steel are used where their technical characteristics are suitable. Daikin has stated that it is pursuing cost reductions by shifting from copper to aluminum, but has not disclosed detailed adoption rates.

Midea. Midea has stated that its domestic air-conditioner products continue to use copper. The company has taken a different approach in overseas markets, where aluminum substitution is more widely accepted and has generated cost benefits. This indicates that Midea's material strategy varies by market rather than following a uniform global approach.

Gree. Gree has stated that aluminum can technically replace copper in certain applications, but currently does not provide equivalent performance and reliability across all relevant conditions. The company therefore has no near-term plan to adopt aluminum substitution in domestic residential products. Gree's position also reflects its ten-year domestic warranty policy and the associated focus on product durability and after-sales costs.

Wanbao. According to our consumer team's channel checks, Wanbao and JD.com launched a 1.5HP zinc-aluminum alloy air conditioner in December 2025. The product was priced at Rmb999, more than 20% below comparable copper-based products, while meeting China's Grade 1 energy-efficiency requirement. The launch demonstrates the potential cost advantage of alternative materials, although broader market acceptance and long-term reliability remain to be established.

Other overseas manufacturers. Lennox has developed aluminum-coil technology for air conditioners, heat pumps and evaporator coils, citing lower weight and reduced corrosion risk. Carrier has applied aluminum coils across its coastal air-conditioning and heat-pump models since 2023, with corrosion resistance cited as a key consideration.

Overall, the largest potential reduction is in heat exchangers, followed by connecting pipes. Progress in compressors, motor windings and other electrical components is likely to be slower because of tighter efficiency, reliability and serviceability requirements. Regional adoption is also likely to remain uneven, with export and selected overseas markets progressing faster than China's domestic residential market.

Autos: fast in the technology while electrification outweighs selective aluminium substitution

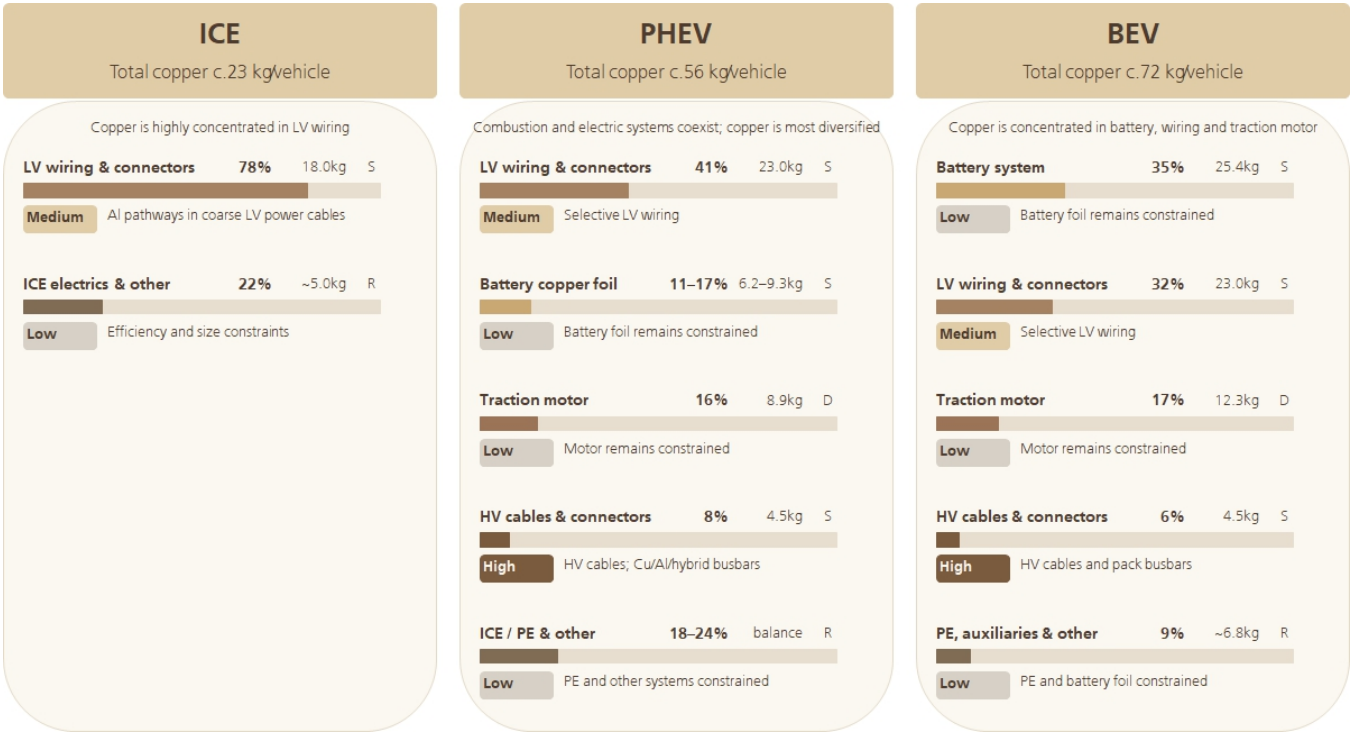
Vehicle electrification remains structurally positive for copper demand, despite declining copper intensity per vehicle. Aluminium substitution is most feasible in low-voltage power cables and battery-pack busbars, with more limited potential in high-voltage harnesses. Battery copper foil, motor windings and compact high-current systems remain more difficult to substitute because of conductivity, thermal, packaging and reliability requirements.

Copper intensity increases materially with vehicle electrification. The representative copper content rises from approximately 23 kg for an internal

combustion engine vehicle to around 56 kg for a plug-in hybrid and 72 kg for a battery electric vehicle. In a representative BEV, the battery system accounts for approximately 35% of copper use, low-voltage wiring and connectors for 32%, the traction motor for 17%, power electronics and auxiliary systems for 9%, and high-voltage cables and connectors for 6%.

Copper intensity is already declining. SMM estimates that BEV copper intensity declined from 76kg per vehicle in 2024 to 72kg in 1H25. PHEV copper intensity declined to 56 kg per vehicle from 58 kg. SMM attributes the decrease to technological improvements and cost optimisation.

Figure 21: Copper density in different auto types: ICE, PHEV, BEV



Source: SMM, UBS estimates

Power cables and battery busbars offer the clearest substitution pathways

Aluminium substitution is most feasible in relatively large power conductors, particularly low-voltage cables, charging harnesses and battery-pack busbars. Aluminium is lighter and cheaper than copper, but its lower conductivity requires a larger conductor cross-section to carry the same current. This trade-off is easier to accommodate in thick power cables and busbars than in small signal wires, compact connectors or tightly packaged electrical components.

Battery busbars provide a particularly practical pathway because OEMs can select copper, aluminium or hybrid copper-aluminium designs according to current, weight, packaging and connection requirements. High-voltage cables also offer substitution potential, although aluminium's higher resistance can increase conductor size and thermal-management requirements. Copper therefore remains preferred where space is limited or where peak current, heat dissipation and terminal reliability are critical.

The transition from 12V to 48V low-voltage architectures could improve the case for aluminium wiring. Higher voltage reduces current at a given power level, which in turn reduces the conductor cross-section required. However, low-voltage harnesses also contain small signal wires, data cables, terminals and connectors that are less suitable for aluminium. We therefore expect adoption to remain component- and platform-specific rather than extending across the entire wiring harness.

Battery copper foil should be assessed separately. Copper remains the standard negative-electrode current collector in mainstream lithium-ion batteries and cannot be

directly replaced by conventional aluminium foil without changing battery chemistry or performance. The more likely sources of copper savings are thinner foil and composite current collectors, rather than direct aluminium substitution.

OEM adoption is progressing, but remains selective

Ferrari. Ferrari introduced aluminum wiring in the 296 model. Combined with smaller cable cross-sections, the change reduced total wiring weight by 15% to 20%. The disclosure demonstrates the weight-saving potential of aluminum conductors, but does not quantify the reduction in copper content per vehicle.

BMW. BMW first used aluminum conductors in the 1 Series in 2011 and subsequently expanded their use in hybrid and battery-electric vehicles. Its sixth-generation eDrive system uses aluminum cables in both high-voltage and low-voltage applications. BMW has not disclosed the resulting copper savings or adoption rate across its vehicle portfolio.

Volkswagen. Volkswagen uses aluminum in selected applications where the material offers specific technical benefits. The company has stated that it has no current plan to replace copper broadly across vehicle wiring. Its approach indicates application-specific material selection rather than a platform-wide transition.

TE Connectivity and Boway Alloy. TE Connectivity and Boway Alloy have developed an aluminum-alloy solution for low-voltage wiring, high-voltage connectors and battery busbars. The companies have focused on electrochemical corrosion and creep performance, which are key constraints in automotive aluminum conductors. Public reporting has referred to potential vehicle applications, but the available primary disclosures reviewed do not establish a verified adoption rate, copper saving per vehicle or cost reduction.

Other automotive manufacturers. Public reporting indicates that Tesla and several Chinese EV manufacturers use aluminum power cables in selected applications. These cases support the technical and commercial feasibility of aluminum conductors, particularly in larger power cables. However, manufacturers have generally not disclosed vehicle-level copper savings or portfolio-wide adoption rates.

Xiaomi. Xiaomi has confirmed that the SU7 uses aluminum conductors in its charging wiring harness. The company stated that the larger cross-sectional area of the aluminum conductor provides equivalent current-carrying capacity, while reducing conductor weight by approximately 30% compared with copper.

Tesla. Tesla introduced aluminum wiring in the Model Y in 2019 and subsequently used the material in the Cybertruck, according to Reuters. Tesla's Model Y electrical documentation also identifies a thick-wall, aluminum-conductor wire specification, confirming that aluminum is used in selected circuits rather than across the entire harness.

Overall, company evidence supports selective substitution rather than broad copper replacement. High-voltage cables and battery busbars have the clearest adoption pathways, while low-voltage wiring remains application-specific. Copper continues to be used where conductivity, thermal performance, packaging and connection reliability are more restrictive.

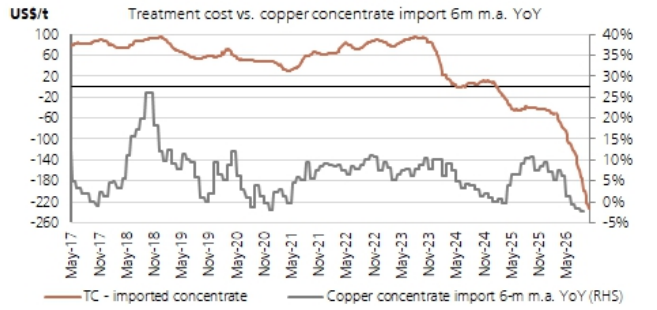
Supply side has been tight, and will be even tighter

Structural constraints limit the supply response. Average ore grades have fallen by roughly 50% over the past two decades, while project lead times, permitting risks and disruptions have increased. Inflation-adjusted copper capex remains around 30% below the previous peak, while the next generation of projects is approximately 65% more capital-intensive. Our global team estimates that around US$175bn of project capex will be required by 2035E to keep supply aligned with demand.

Copper prices have remained resilient at elevated levels despite geopolitical tensions in the Middle East and rising interest-rate expectations. China's cumulative concentrate imports have turned negative on a YoY basis since April, marking the first contraction since 2021, while spot treatment charges have fallen below negative US$200/t, highlighting severe concentrate shortages. Concentrate inventories at major Chinese ports have more than halved from their peak, prompting smelters to slow production. As a result, China's refined copper output could begin to decline in July, while scrap imports offer limited scope to offset the shortfall.

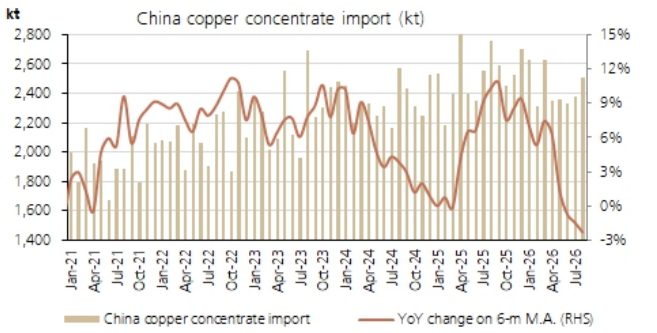
These constraints are feeding through to the refined market. China's copper inventories have fallen to record lows, while global inventories continue to decline, leaving limited buffer against further supply disruptions or stronger-than-expected demand.

Figure 22: Continued TC/RC declining



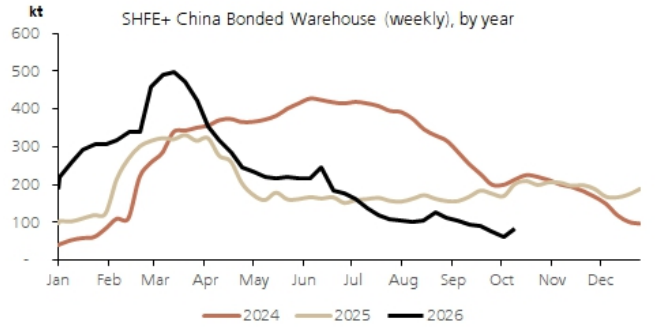
Source: SMM

Figure 23: China's copper concentrate imports continued to fall



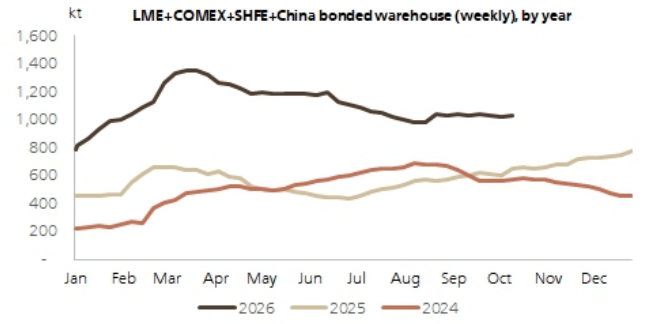
Source: SMM, Wind

Figure 24: China inventory near record low



Source: SMM, Wind, UBS estimates

Figure 25: Global inventory keeps going down



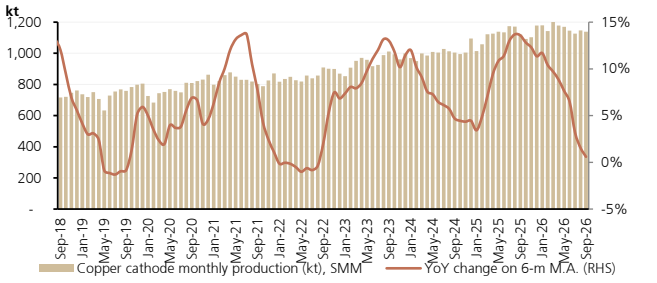
Source: SMM, Wind, UBS estimates

Figure 26: China concentrate inventory halved from peak



Source: SMM, Wind, UBS estimates

Figure 27: China smelters slow down



Source: SMM, Wind, UBS estimates

Figure 28: China SD model

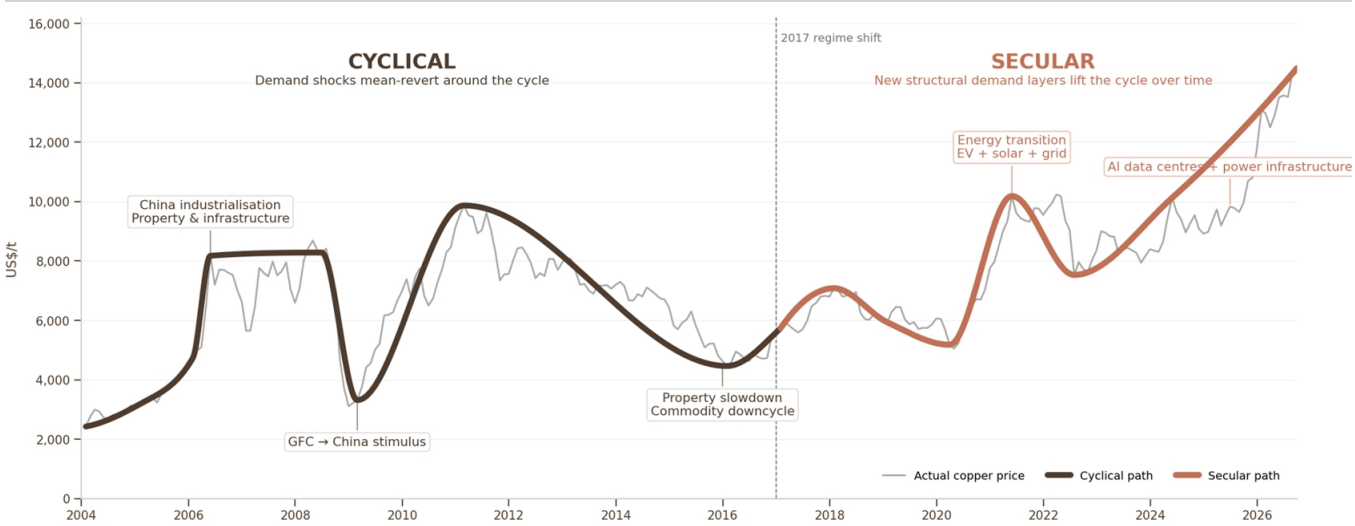
Unit: Cu, kt	2022	2023	2024	2025	2026E	2027E
China supply						
China mine production	1,913	1,743	1,794	1,803	1,906	1,983
Concentrate import	6,420	6,910	7,029	7,590	7,363	7,289
China total mine supply	**8,333**	**8,653**	**8,823**	**9,394**	**9,269**	**9,272**
Scrap to smelter	1,781	2,117	2,233	2,375	2,718	2,820
China smelter output	9,590	10,452	10,520	10,897	11,251	11,251
Blister/anode import	1,155	992	887	746	769	761
Scrap to refiner	450	629	722	750	1,144	1,187
China refiner output	**10,481**	**11,488**	**11,888**	**12,311**	**12,311**	**12,311**
Unwrought copper import	3,637	3,458	3,583	3,164	2,689	2,662
SXEW	39	40	38	38	38	38
Total supply	**14,157**	**14,986**	**15,509**	**15,512**	**15,038**	**15,011**
YoY	3%	6%	3%	0%	-3%	0%
China demand						
1. Construction	**2,591**	**2,513**	**2,401**	**2,257**	**2,190**	**2,124**
% in total demand	17%	15%	14%	13%	12%	12%
YoY	-7%	-3%	-4%	-6%	-3%	-3%
2. Total Transport	**1,326**	**1,460**	**1,614**	**1,775**	**1,889**	**2,046**
% in total demand	9%	9%	9%	10%	11%	11%
YoY	14%	10%	11%	10%	6%	8%
3. Renewables	**722**	**1,540**	**1,844**	**2,245**	**1,664**	**1,747**
% in total demand	5%	9%	11%	13%	9%	10%
YoY	1%	113%	20%	22%	-26%	5%
4. Power Grid	**5,298**	**5,411**	**5,528**	**5,569**	**5,890**	**6,156**
% in total demand	34%	33%	32%	31%	33%	34%
YoY	5%	2%	2%	1%	6%	5%
5. Appliances	**2,724**	**2,962**	**3,148**	**3,242**	**3,210**	**3,210**
% in total demand	18%	18%	18%	18%	18%	18%
YoY	0%	9%	6%	3%	-1%	0%
6. Machinery & Equipment	**1,892**	**1,896**	**1,949**	**2,007**	**2,168**	**2,211**
% in total demand	12%	11%	11%	11%	12%	12%
YoY	-1%	0%	3%	3%	8%	2%
7. Others	**930**	**819**	**824**	**791**	**815**	**839**
% in total demand	6%	5%	5%	4%	5%	5%
YoY	-5%	-12%	1%	-4%	3%	3%
Total demand	**15,483**	**16,600**	**17,308**	**17,887**	**17,826**	**18,333**
YoY	1%	7%	4%	3%	0%	3%
Direct use scrap	**1,834**	**1,966**	**2,032**	**2,002**	**1,838**	**1,906**
Demand for refined copper	**13,608**	**14,634**	**15,276**	**15,885**	**15,988**	**16,427**
YoY	1%	8%	4%	4%	1%	3%
China refined copper balance	409	351	234	-373	-950	-1,416
Balance, rebased on 2023 chg	**-27**	**-85**	**-203**	**-810**	**-1387**	**-1852**

Source: UBS estimates

Secular copper growth calls for a re-rating of HK/ China miners

Copper has historically traded through pronounced investment and inventory cycles, with higher prices eventually encouraging new supply and slowing demand. We believe the current cycle is different. Electrification, grid investment and AI infrastructure are broadening the sources of demand, while declining ore grades, longer development timelines and higher capital intensity are limiting the supply response. If these conditions persist, copper prices could remain higher for longer, with US$13,000/t becoming a new norm rather than a short-lived cyclical peak.This distinction matters for equity valuation. A traditional cyclical framework assumes that stronger commodity prices and miner earnings will mean-revert rapidly. A structural framework instead supports a longer period of elevated margins, cash generation and returns. Overseas copper miners have increasingly been valued on the latter basis, but HK/China miners continue to trade largely as traditional cyclicals.

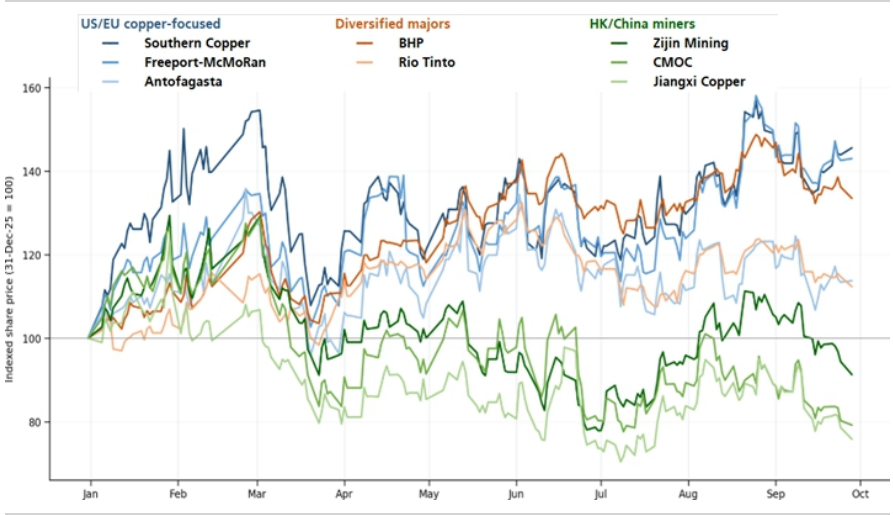
Figure 29: Copper prices: from cyclical to secular



Source: Wind, UBS estimates

Figure 30 also suggests that the rerating of copper equities has begun with US/Europe names and then broadened to Australian miners, while HK/China copper names underperform. We believe Chinese copper names could be next, as the market increasingly prices in a higher-for-longer copper environment, with US$13,000–15,000/t potentially becoming the new normal. Sustained demand from AI data centres should support copper consumption, while supply constraints are unlikely to ease quickly. Rising interest from overseas funds in China could provide an additional catalyst for a valuation catch-up.

Figure 30: Copper stocks: 2026 YTD share price performance

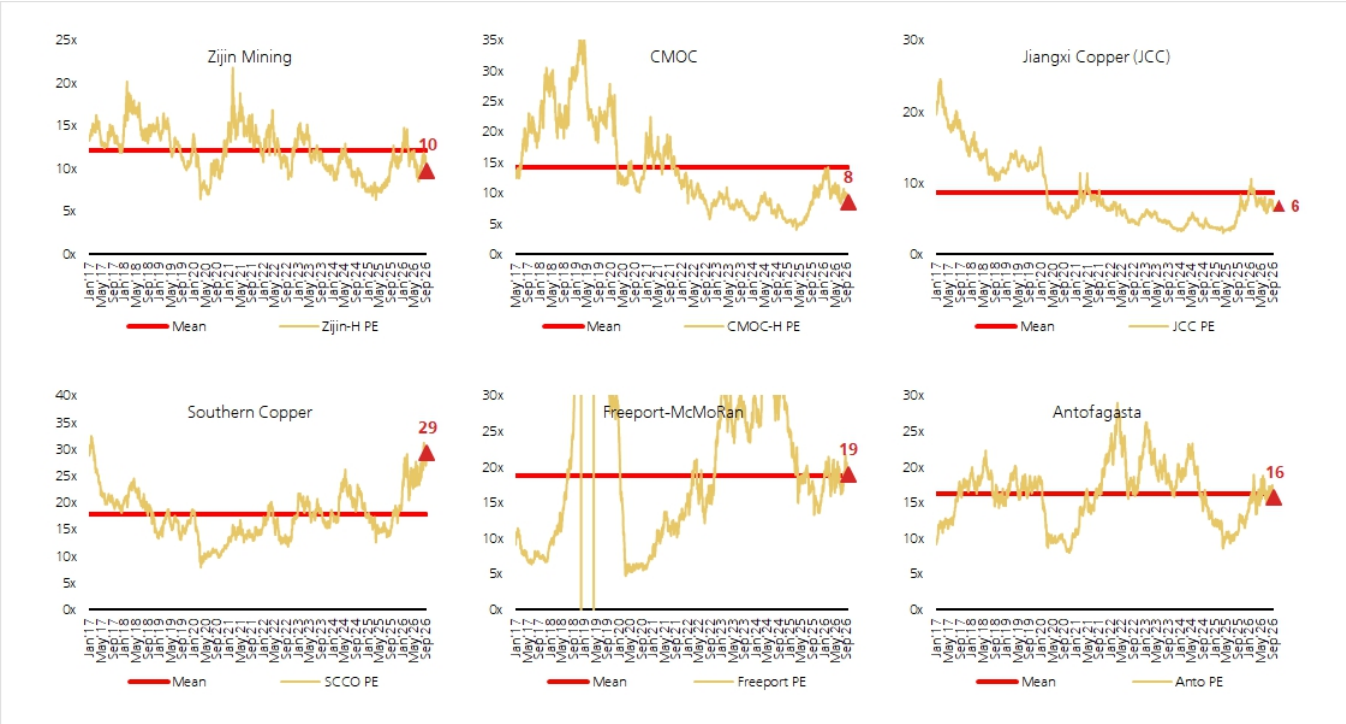


Source: Wind, UBS estimates

The valuation gap remains wide among overseas copper miners and HK/China copper miners

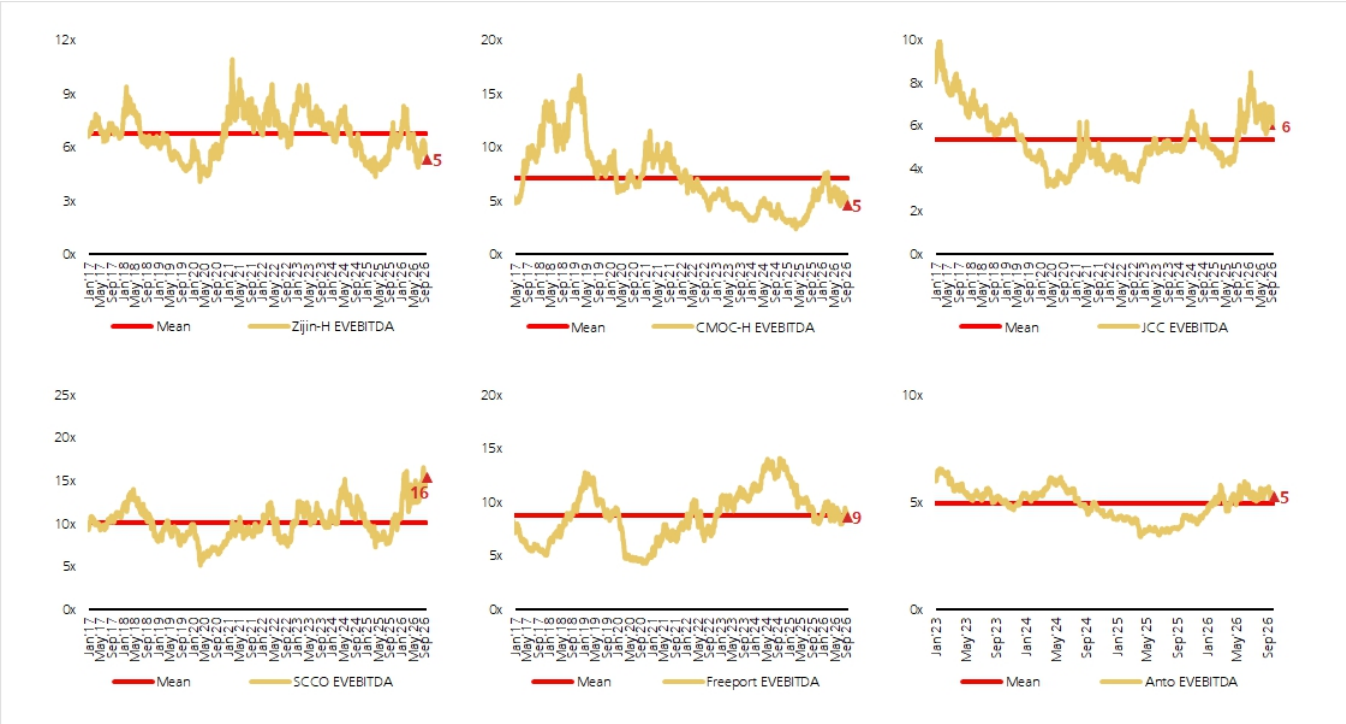
The historical valuation charts reinforce this divergence. Overseas names, particularly Southern Copper, have experienced a sustained multiple expansion as investors attached greater value to long-life resources, low-cost production and copper scarcity. By contrast, the valuation multiples of Zijin Mining, CMOC and Jiangxi Copper remain closer to their historical cyclical ranges, even as their operating performance, resource bases and production outlooks have improved. HK/China copper miners currently trade at average 8x 12-month forward P/E, versus 15–30x for overseas peers. The gap is also significant on EV/EBITDA, with HK/China miners trading at around 6x 12-month forward EV/EBITDA, compared with an average of 10x overseas. In our view, this discount reflects an outdated cyclical valuation framework and should narrow as the market gains confidence in a higher long-term copper price and more durable earnings.

Figure 31: China and overseas copper miners valuation changes over time - PE



Source: UBS estimates

Figure 32: China and overseas copper miners valuation changes over time - EV/EBITDA



Source: UBS estimates

Volume growth is scarce, but Zijin and CMOC stand out

Our comparison of major global miners shows that the supply response remains uneven, with only a limited group offering both meaningful volume growth and competitive costs.

Across the major miners in our analysis, several established producers are expected to deliver only moderate growth through 2030. Southern Copper's output is forecast to rise from 956kt in 2025 to 1,014kt in 2030, implying a 2% CAGR from 2026 to 2030. Rio Tinto is also expected to deliver relatively modest growth, while BHP's production is forecast to remain below its 2025 level by 2030. This reinforces the broader point that higher prices do not automatically generate a rapid supply response.

By contrast, Zijin and CMOC offer comparatively visible growth:

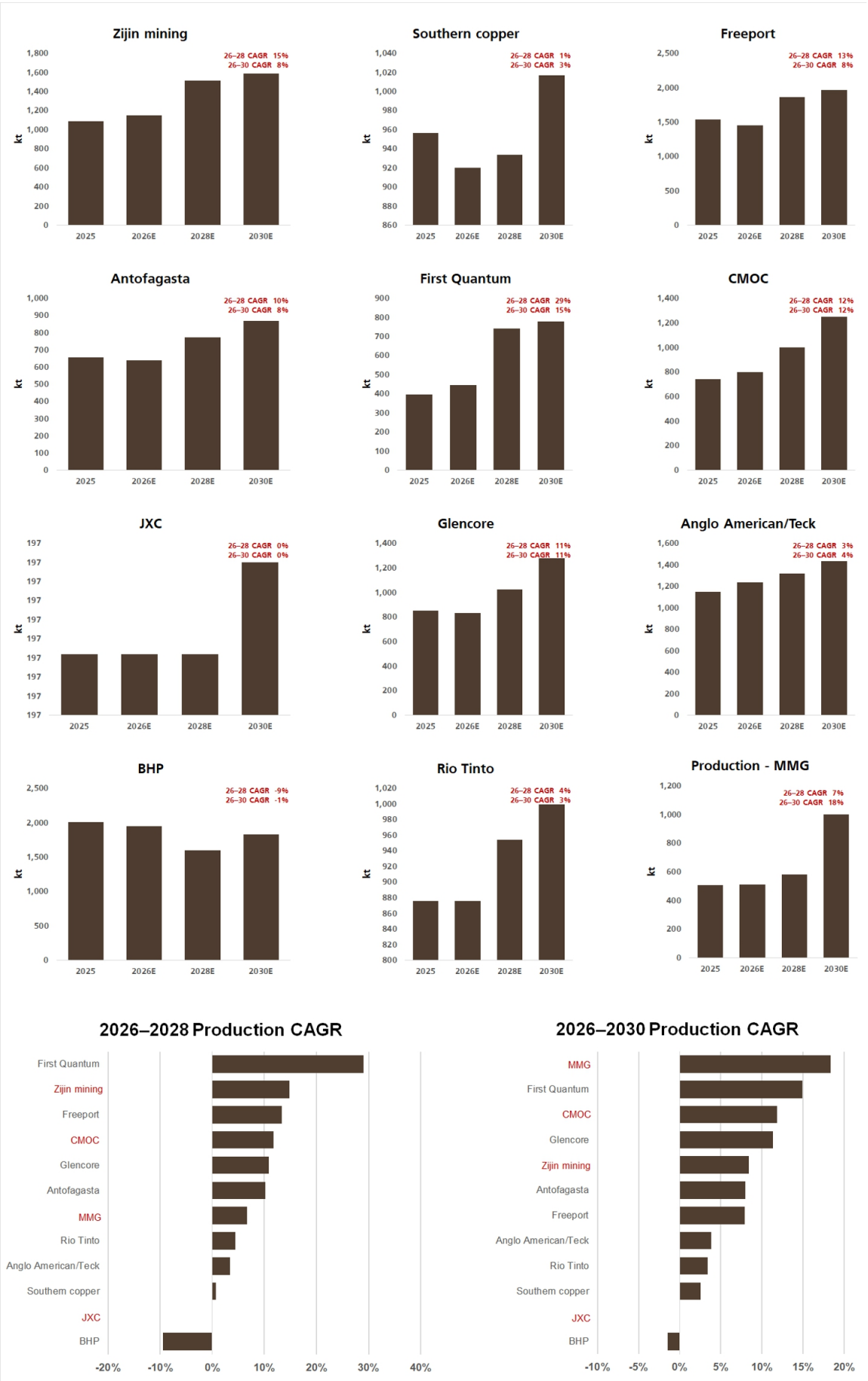
- Zijin Mining: production rises from 1.1mt in 2025 to 1.5mt in 2028E and 1.6mt in 2030E, implying a 15% CAGR in 2026–28E and 8% in 2026–30E.
- CMOC: production increases from 740kt in 2025 to 1mt in 2028E and 1.25mt in 2030E, representing a 12% CAGR over both 2026–28E and 2026–30E.
- Jiangxi Copper: attributable mine production remains broadly stable at approximately 200kt, but all its copper mines are located domestically, suggesting stable cost. Besides, Solgold should drive medium-long term growth after 2030.

Figure 33: Major miners next 3-5 year mining plan

Consolidated Copper Production (kt)								
	Miner types	2025	2026E	2028E	2030E	26-28 CAGR	26-30 CAGR	Mark
Zijin mining	copper	1,085	1,148	1512	1585	15%	8%	UBSe
Southern copper	copper	956	920	934	1017	1%	3%	UBSe
Freeport	copper	1,535	1,449	1862	1963	13%	8%	UBSe
Antofagasta	copper	654	637	773	866	10%	8%	UBSe
First Quantum	copper	396	446	742	777	29%	15%	UBSe
CMOC	copper	741	800	1000	1250	12%	12%	UBSe
Glencore	diversified	852	832	1023	1279	11%	11%	UBSe
Anglo American/Teck	diversified	1,149	1,232	1316	1433	3%	4%	UBSe, Pro-forma number
BHP	diversified	2,007	1,943	1592	1829	-9%	-1%	UBSe
Rio Tinto	diversified	875	875	954	999	4%	3%	UBSe
MMG	copper	507	510	580	1000	7%	18%	Company + UBSe
JXC	copper + smelting	197	197	197	197	0%	0%	UBSe

Source: Company data, UBS estimates

Figure 34: Major copper miners volume CAGR: from 2026-2028-2030

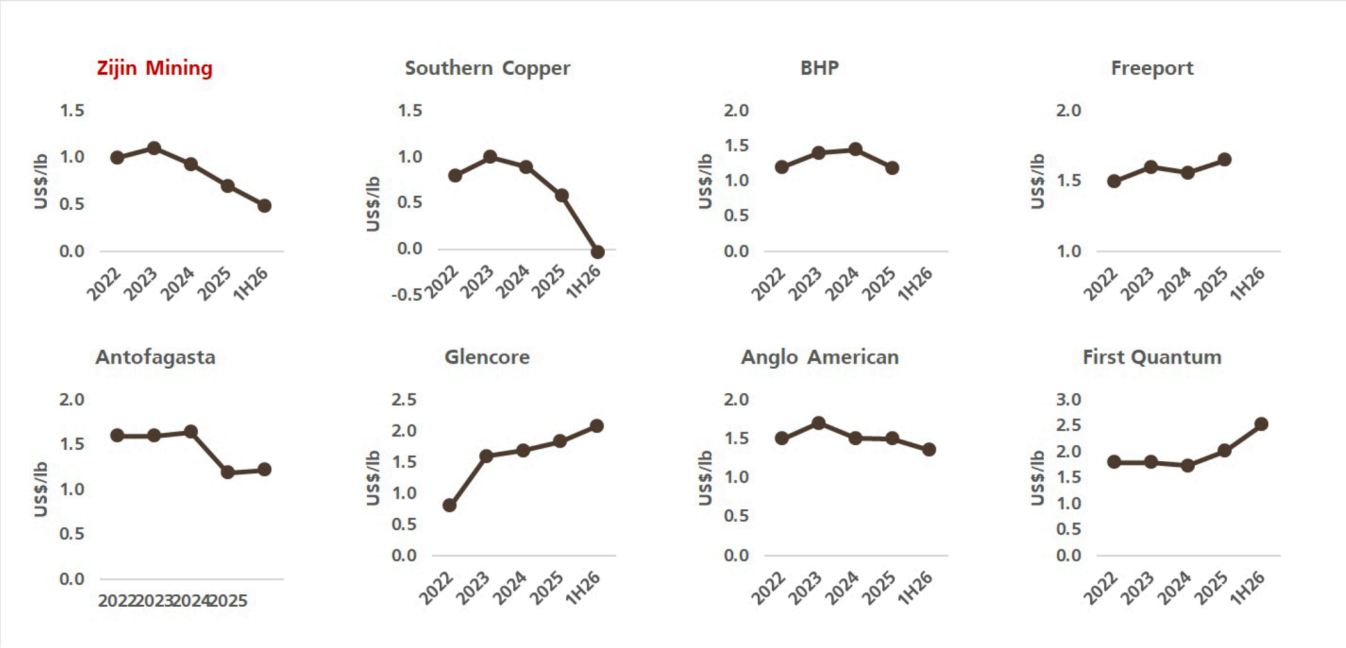


Source: Company data, UBS estimates

Cost position determines the quality of copper-price exposure

Production growth should be assessed together with cost position. Higher-cost miners may generate significant earnings upside during a strong copper market, but they also face greater margin volatility if prices or operating conditions weaken. Lower-cost miners can retain stronger cash generation through the cycle and are therefore more likely to receive a higher structural valuation. The cost curves show substantial differences across the peer group. Zijin remains competitively positioned, with a diversified asset base and cost support from by-product exposure. CMOC also benefits from the quality and grade of its African copper assets, although logistics, regional operating conditions and fiscal mechanisms remain important variables. For several overseas peers, declining grades and operating disruptions are contributing to higher or less predictable costs.

Figure 35: Major copper miners C1 cost trajectory over the year



Source: Company data

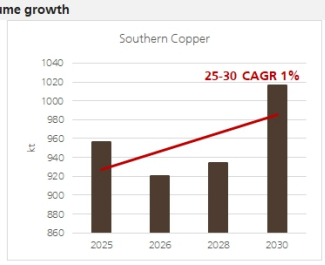
Zijin Mining: Southern Copper-like quality, with stronger growth and a lower valuation

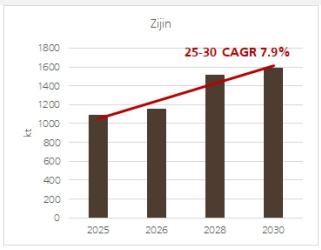
Zijin Mining combines visible volume growth, strong execution and a consistent track record of earnings delivery, in our view. As China's largest copper miner, we believe Zijin offers a comparable resource base and cost position to Southern Copper, but with stronger production growth and a more undemanding valuation. **Stronger production growth**. Zijin is expected to deliver a 15% copper production CAGR during 2026–28E, compared with approximately 2% for Southern Copper. By 2030, Zijin's copper production could reach approximately 1.6Mt, considerably above Southern Copper's forecast output of around 1.0Mt. **Larger and more diversified earnings base**. In addition to copper, Zijin has meaningful exposure to gold and lithium. Gold provides earnings diversification and potential downside protection during periods of macro uncertainty, while lithium offers longer-term optionality. The more diversified portfolio can create additional earnings variables, but it also reduces dependence on a single commodity. **Undemanding resource valuation.** Zijin screens attractively on both EV/resources and EV/reserves, suggesting the market assigns limited value to its underlying mineral endowment and development pipeline. The discount is particularly notable given its growth rate and execution record.

Figure 36: Comparison of Zijin Mining and Southern Copper

Comparison of Zijin Mining vs. Southern Copper

	Southern Copper (SCCO.N)	Zijin Mining (601899.SS/2899.HK)
Equity basis reserves/resources (2025)		
Resources	12,863	8,621
Reserves	4,942	4,463
EV/Resources	14.0	14.6
EV/ Reserves	36.3	28.2
Volume growth		



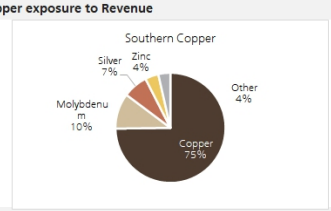


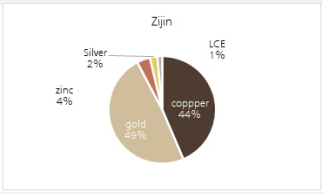
Cost trajectory





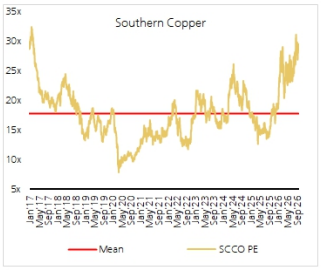
Copper exposure to Revenue

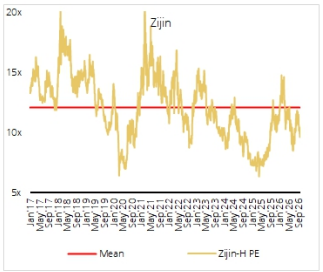




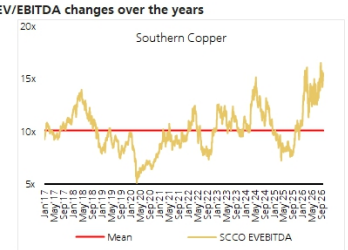
Valuation pattern changes

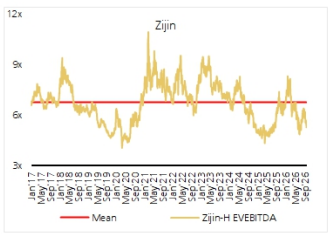
PE changes over the years



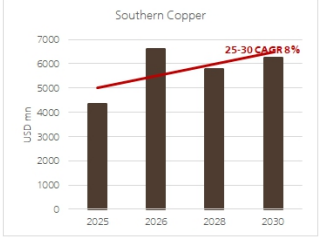


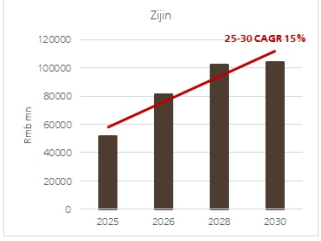
EV/EBITDA changes over the years





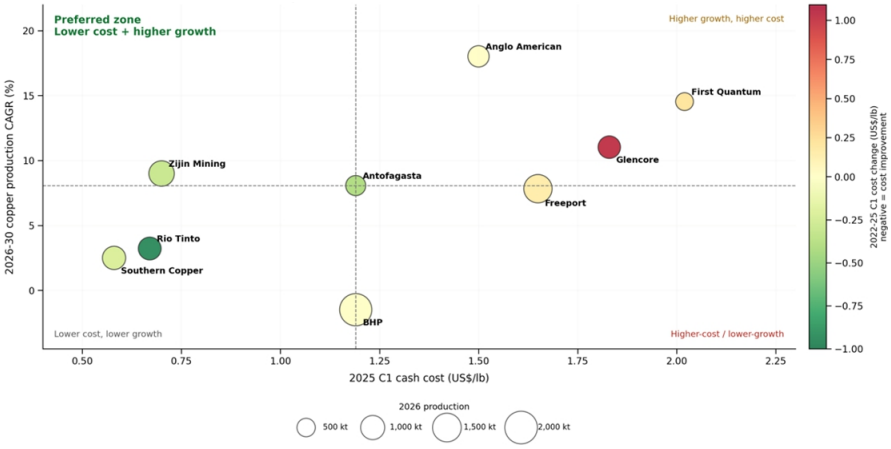
Company net earnings growth





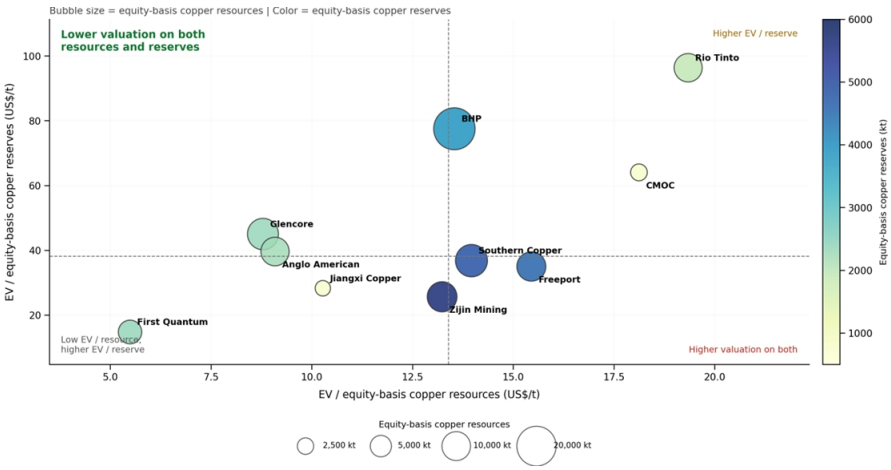
Source: Company data, UBS estimates

Figure 37: Zijin combines above-average production growth with competitive costs



Source: Company data, UBS estimates

Figure 38: Zijin screens attractively on EV/reserves and EV/resources



Source: Company data, UBS estimates

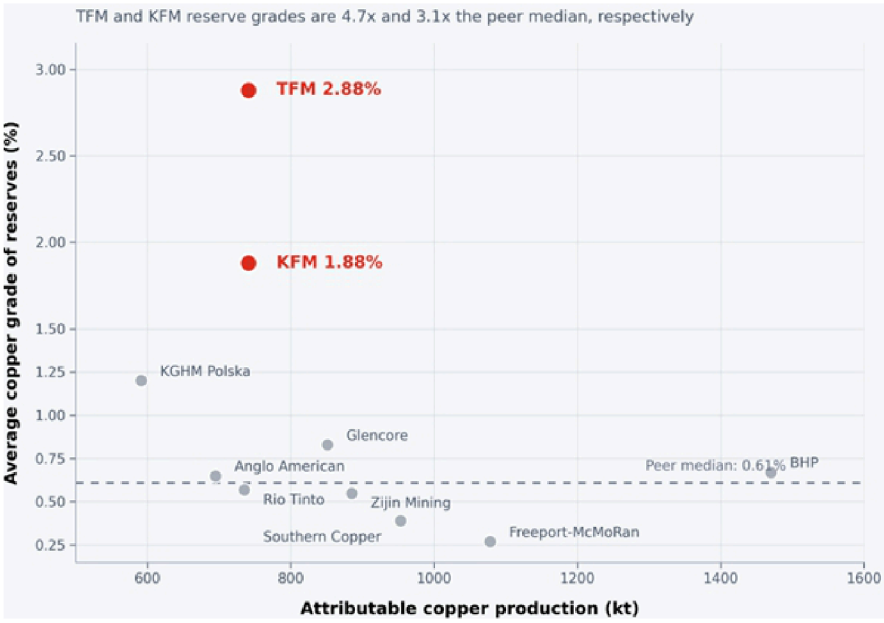
CMOC: high-grade resources support volume growth and resilient margins

We believe CMOC offers a different but similarly compelling investment case. Its principal advantage is the combination of abundant, high-grade resources and visible production growth. Copper output is forecast to increase from 741kt in 2025 to 1.0Mt in 2028 and 1.25Mt in 2030, implying a 12% CAGR over 2026–30E.

Recent production has exceeded guidance, strengthening confidence in the company's ability to deliver volume. This is important because the market has historically applied execution and jurisdictional discounts to DRC mining operations. Consistent delivery can reduce these discounts over time.

High ore grades support a competitive operating position, while we expect all-in costs to remain at c.US$5,200–5,700/t. CMOC's cost position provides meaningful leverage to a higher copper price while also supporting margins under less favourable scenarios. Although disruptions in the Middle East may affect logistics and input costs, we see the overall impact as manageable within the current cost outlook.

Figure 39: CMOC's owns high-grade reserves stand out



Copper company	Attrib. prod. (kt)	Avg reserve grade
CMOC Group	741	TFM: 2.88% KFM: 1.88%
KGHM Polska	591	1.20%
Glencore	851	0.83%
BHP	1,470	0.67%
Anglo American	695	0.65%
Rio Tinto	735	0.57%
Zijin Mining	885	0.55%
Southern Copper	953	0.39%
Freeport-McMoRan	1,078	0.27%

Table sorted by average reserve grade, high to low.

Source: Company data, UBS estimates

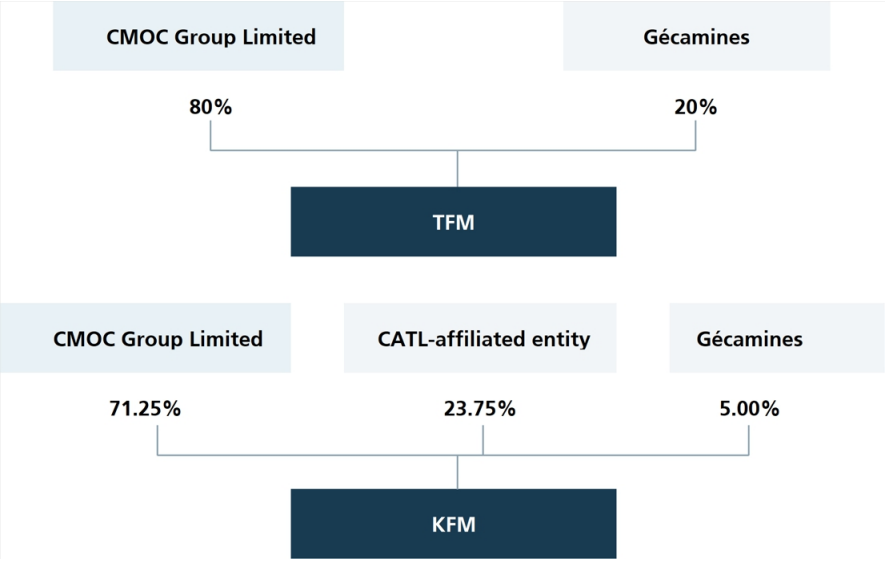
Despite CMOC's strong volume growth and high-grade resources, the DRC's country risks cannot be overlooked. The DRC's 2018 Mining Code requires Congolese nationals to hold at least 10% of the share capital of mining companies holding exploitation permits, with 5% reserved for Congolese employees and 5% for other Congolese individuals. This requirement is separate from, and additional to, the Congolese state's existing 10% free-carried interest.

Although the requirement was introduced in 2018, it remained largely unenforced for approximately eight years due to the absence of detailed implementation rules. The government began pushing for its implementation in 2026. As of the date of this report, we have not seen public confirmation that the implementing decree has been formally signed. Nevertheless, we cannot rule out its eventual implementation.

Three key issues remain unresolved. First, the draft does not specify whether the shares should be transferred at market value, book value or a negotiated discount. Second, it remains unclear whether existing shareholders would be required to transfer part of their holdings and whether the requirement would apply retrospectively to existing projects. Third, it is uncertain whether the stakes already held by Congolese state-owned shareholders, including Gécamines' 20% interest in TFM and the Congolese state shareholder's 5% interest in KFM, could be counted towards the 10% local ownership requirement.

Given these uncertainties and the potential dilution of existing shareholders' economic interests, we apply a 10% discount to our target multiple to reflect the associated country and regulatory risks.

Figure 40: TFM/KFM shareholder structure



Source: Company data

Figure 41: DRC mining law and citizen shareholding timeline

2002	**Old Mining Code**	The state receives a 5% free-carried interest. This provision underpins a 5% state stake in Kamoa.
March 2018	**New Mining Code enacted**	Art. 71 bis requires 10% citizen ownership. Art. 144 assigns 5% to Congolese individuals able to buy shares and 5% to Congolese employees. The 10-year stabilization period ends. The new rules apply immediately to all miners.
2018–2025	**Provision not implemented**	No implementing rules were issued. No company complied.
Jan 30, 2026	**Ministry seeks evidence**	The Mining Ministry asks for proof of the 5% employee stake. Noncompliant companies receive a grace period until July 31, 2026.
Jun 11, 2026	**Industry requests extension**	CMOC, Glencore, Ivanhoe and ERG coordinate through the chamber. They question whether existing shareholders must cede stakes and whether the rule applies to older projects.
Jun 23, 2026	**Draft decree reported**	Reuters reports employee ownership via a cooperative. Up to 80% of dividends would repay an interest-free company loan. Capital increases could not dilute the 10% stake.
Jul 21, 2026	**Government and chamber agree**	Both sides agree to sign after minor technical changes and establish an ad hoc committee. The draft gives companies six months to comply.
Jul 31, 2026	**Grace period expires**	The obligation takes effect. The joint committee still seeks a decree on pricing, financing and the shareholding vehicle. Rules on allocation, valuation and payment remain unpublished.
Aug 2026 onward	**Talks continue**	Negotiations continue.

Source: Reuters, Mining.com

Figure 42: Government negotiations in mining: case comparison

Case	Government demand	Settlement terms	Financial and market impact
CMOC · TFM (2022–23)	Gécamines alleged understated reserves and claimed US$7.6bn.	Ownership unchanged. CMOC agreed to pay US$800m over 2023–28 and guarantee at least US$1.2bn in dividends.	H1 2023 attributable net profit was expected to fall by over 80%. Full-year 2023 attributable net profit still rose 36%.
Ivanhoe / Zijin · Kamoa (2016)	Increase the state stake from 5% to 20%.	Transfer a 15% stake for nominal consideration.	Secured priority repayment of shareholder loans and a life-of-project tax and fiscal stability commitment. Ivanhoe's interest was diluted by about 16%.
Zimbabwe · Zimplats (2013)	51% local ownership, including 10% each for employees and the community.	Nominally equal-value transfer priced at US$971m. The company provided 10% interest vendor financing, to be repaid with 85% of dividends.	Implementation later faltered, with a dispute over unpaid amounts for the community stake.
Tanzania · Acacia (2017)	Banned concentrate exports and issued a US$190bn tax bill.	16% free-carried state interest, a US$300m payment and a 50/50 split of economic benefits.	2017 EBITDA fell 38%. The company booked a US$644m impairment, and its share price fell 66%.
Mali · Barrick (2023–25)	Increase the state share under the new Mining Code.	About US$430m in cash and acceptance of the new Mining Code.	Mine operations stopped. Barrick recognized a US$1.035bn deconsolidation loss.
Indonesia · Freeport (2018)	Divest 51% of the Grasberg project.	Government-side buyers paid approximately US$3.85bn in cash.	The ownership interest roughly halved in exchange for an extension of mining rights.

Source: Company data, UBS

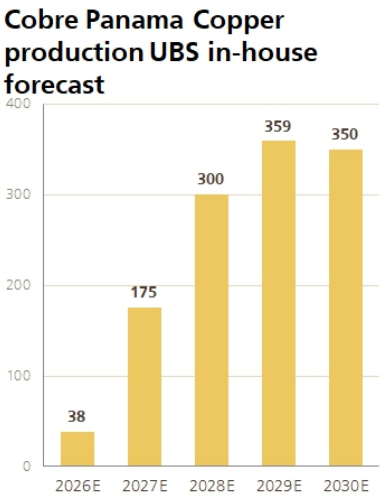
Jiangxi Copper: stable domestic assets with multiple sources of optionality

Jiangxi Copper's investment case differs from those of Zijin and CMOC. Its attributable mine-production profile is relatively stable, so the company offers less direct organic volume growth. However, it combines established domestic copper assets and smelting capacity with several potential earnings catalysts that we think are not fully captured by its current valuation.

The most important nearer-term catalyst is Jiangxi Copper's 18% interest in **First Quantum**. A restart of Cobre Panamá could materially increase First Quantum's earnings contribution, while the Kansanshi S3 expansion provides an additional source of potential growth.

Figure 43: UBS in-house estimates of Cobre Panama restart progress

Nov 2023	**Court ruling**	Panama's Supreme Court voids the concession law, initiating the mine closure.
Jun 18, 2026	**SGS audit published**	The integral audit reports an 88% compliance score. UBS views it as the basis for a government decision.
2026E	**2026 output forecast**	UBS forecasts 38 kt of copper production for full-year 2026.
Late 2026	**Commission reports**	UBS expects the Ministerial Commission to submit recommendations to President Mulino by late 2026.
~Dec 31, 2026	**Restart accord modeled**	UBS models a bilateral restart agreement around year-end as a key catalyst.
H1 2027	**Commercial restart**	UBS expects a full commercial restart after pit mining resumes.
2027E	**New concession terms**	UBS assumes an effective tax rate of about 45% and a 10% Panamanian state equity stake from 2027.
End-2027	**Ramp-up**	UBS expects the mine to reach 175 kt of copper in 2027.
2028E	**Near normalized operations**	UBS forecasts 300 kt of copper production as operations normalize.



Source: UBS estimates

Looking further ahead, **SolGold** represents a source of strategic optionality rather than a near-term earnings driver, as production is not expected to commence until after 2030 based on the company's current guidance. Its ultimate value will depend on project financing, permitting progress, the construction schedule and the eventual ownership structure. Under the latest development plan, the project could deliver annual copper production of approximately 100kt, with further upside potential.

Jiangxi Copper also offers incremental upside through its **copper foil** and **tungsten** businesses. Its 31% interest in listed tungsten subsidiary Jiaxin International provides exposure to tightening tungsten fundamentals, with additional strategic upside potential should Jiangxi Copper increase its ownership stake in the future. Meanwhile, the copper foil business offers downstream growth exposure to batteries and advanced manufacturing. The business generated c. Rmb50mn of earnings in 1H26, and management expects it to contribute an additional Rmb200mn to Rmb300mn to earnings in 2027. Together, these businesses diversify Jiangxi Copper's earnings base beyond its core mining and smelting operations.

Figure 44: Volume & earnings comparison for Zijin, CMOC, Jiangxi Copper

	Consolidated Copper Production Volume							Earnings					
	Miner types	2025	2026E	2028E	2030E	26-28 CAGR	26-30 CAGR	2025	2026E	2028E	2030E	26-28 CAGR	26-30 CAGR
Zijin mining	copper	1,085	1,148	1,512	1,585	15%	8%	51,777	81,414	102,263	104,351	12%	6%
CMOC	copper	741	800	1,000	1,250	12%	12%	20,339	32,186	38,943	46,888	10%	10%
JXC	copper + smelting	197	197	197	197	0%	0%	10,818	16,081	18,282	17,365	7%	2%

Source: UBS estimates

Sensitivity analysis for Zijin mining, CMOC and Jiangxi Copper 2027E earnings and PE

Figure 45: Zijin 2027E earnings and PE sensitivity analysis

Zijin mining						
Gold \ Copper	10,000	12,000	13,000	14,000	15,000	20,000
3,500	57,328	65,984	70,312	74,639	78,967	100,606
4,000	64,074	72,729	77,057	81,385	85,713	107,351
4,500	70,819	79,475	83,803	88,130	92,458	114,097
5,000	77,565	86,220	90,548	94,876	99,204	120,842
5,500	84,310	92,966	97,294	101,621	105,949	127,588

Implied 2027E PE (H share)						
Gold \ Copper	10,000	12,000	13,000	14,000	15,000	20,000
3,500	13.1	11.4	10.7	10.1	9.5	7.5
4,000	11.7	10.3	9.8	9.2	8.8	7.0
4,500	10.6	9.5	9.0	8.5	8.1	6.6
5,000	9.7	8.7	8.3	7.9	7.6	6.2
5,500	8.9	8.1	7.7	7.4	7.1	5.9

Implied 2027E PE (A share)						
Gold \ Copper	10,000	12,000	13,000	14,000	15,000	20,000
3,500	14.3	12.4	11.6	11.0	10.4	8.1
4,000	12.8	11.3	10.6	10.1	9.6	7.6
4,500	11.6	10.3	9.8	9.3	8.9	7.2
5,000	10.6	9.5	9.0	8.6	8.3	6.8
5,500	9.7	8.8	8.4	8.1	7.7	6.4

Source: UBS estimates, as of Oct 13th

Figure 46: CMOC 2027E earnings and PE sensitivity analysis

CMOC						
Gold \ Copper	10,000	12,000	13,000	14,000	15,000	20,000
3,500	22,615	30,036	31,104	33,046	34,988	44,696
4,000	23,118	30,538	31,607	33,548	35,490	45,199
4,500	23,620	31,041	32,109	34,051	35,993	45,701
5,000	24,123	31,543	32,612	34,553	36,495	46,203
5,500	24,625	32,046	33,114	35,056	36,998	46,706

Implied 2027E PE (H share)						
Gold \ Copper	10,000	12,000	13,000	14,000	15,000	20,000
3,500	12.5	9.4	9.1	8.6	8.1	6.3
4,000	12.2	9.3	8.9	8.4	8.0	6.3
4,500	12.0	9.1	8.8	8.3	7.9	6.2
5,000	11.7	9.0	8.7	8.2	7.8	6.1
5,500	11.5	8.8	8.5	8.1	7.6	6.1

Implied 2027E PE (A share)						
Gold \ Copper	10,000	12,000	13,000	14,000	15,000	20,000
3,500	16.3	12.3	11.8	11.1	10.5	8.2
4,000	15.9	12.1	11.7	11.0	10.4	8.2
4,500	15.6	11.9	11.5	10.8	10.2	8.1
5,000	15.3	11.7	11.3	10.7	10.1	8.0
5,500	15.0	11.5	11.1	10.5	10.0	7.9

Source: UBS estimates, as of Oct 13th

Figure 47: Jiangxi Copper 2027E earnings and PE sensitivity analysis

JCC						
Gold \ Copper	10,000	12,000	13,000	14,000	15,000	20,000
3,500	12,480	14,225	15,097	15,969	16,841	21,202
4,000	12,827	14,571	15,444	16,316	17,188	21,549
4,500	13,174	14,918	15,790	16,663	17,535	21,896
5,000	13,521	15,265	16,137	17,009	17,882	22,242
5,500	13,867	15,612	16,484	17,356	18,228	22,589

Implied 2027E PE (H share)						
Gold \ Copper	10,000	12,000	13,000	14,000	15,000	20,000
3,500	7.9	6.9	6.5	6.2	5.9	4.7
4,000	7.7	6.8	6.4	6.0	5.7	4.6
4,500	7.5	6.6	6.3	5.9	5.6	4.5
5,000	7.3	6.5	6.1	5.8	5.5	4.4
5,500	7.1	6.3	6.0	5.7	5.4	4.4

Implied 2027E PE (A share)						
Gold \ Copper	10,000	12,000	13,000	14,000	15,000	20,000
3,500	11.8	10.3	9.8	9.2	8.7	6.9
4,000	11.5	10.1	9.5	9.0	8.6	6.8
4,500	11.2	9.9	9.3	8.8	8.4	6.7
5,000	10.9	9.6	9.1	8.7	8.2	6.6
5,500	10.6	9.4	8.9	8.5	8.1	6.5

Source: UBS estimates, as of Oct 13th

Figure 48: Copper comps table

	Ticker	Rating	Copper/ Diversified	Price	PT	Market cap	PE (x)		EV/EBITDA (x)		PB (x)		ROE		Div. yield	
				(LC)	(LC)	US$ m	2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E
Copper																
Zijin A	601899.SS	Buy		30.40	50.00	120,702	9.9	8.6	6.5	5.1	3.2	2.6	37.3	33.4	4.3	5.3
CMOC A	603993.SS	Buy		17.04	29.00	54,984	11.3	10.4	6.1	5.4	3.4	2.8	33.7	28.9	2.7	2.9
Jiangxi A*	600362.SS	Not Rated		41.55	NA	21,494	9.1	9.6	9.0	9.4	1.5	1.4	17.5	14.8	4.9	4.9
Western*	601168.SS	Not Rated		35.31	NA	12,570	12.0	10.5	8.2	7.8	3.4	2.7	30.3	26.6	2.0	2.8
Jinchengxin*	603979.SS	Not Rated		72.52	NA	6,523	13.5	10.0	6.7	6.1	3.1	2.5	25.3	24.7	1.4	1.5
Yunnan*	000878.SZ	Not Rated		15.69	NA	3,320	9.4	7.5	NA	NA	2.1	1.8	22.6	24.1	4.2	5.3
Tongling*	000630.SZ	Not Rated		5.73	NA	1,217	11.4	7.5	NA	NA	1.9	1.6	17.3	22.9	4.6	6.9
A-share avg.							**11.3**	**9.4**	**7.3**	**6.8**	**3.0**	**2.5**	**27.7**	**26.3**	**3.4**	**4.2**
Zijin H	2899.HK	Buy		32.90	56.00	111,420	9.3	8.0	6.1	4.8	3.0	2.4	37.3	33.4	4.6	5.7
CMOC H	3993.HK	Buy		15.58	32.00	42,682	9.0	8.1	4.7	4.2	2.6	2.1	33.7	28.9	3.4	3.7
JCC H	0358.HK	Buy		33.04	65.00	14,578	6.2	5.7	5.8	5.4	1.0	0.9	18.4	17.2	5.7	6.2
MMG*	1208.HK	Not Rated		8.67	NA	14,192	7.9	7.5	6.3	6.0	2.0	1.6	31.4	24.4	0.1	-
China Nonferrous*	1258.HK	Not Rated		15.48	NA	6,882	9.5	9.3	4.0	4.1	2.3	1.9	26.9	22.5	3.7	3.9
China Gold Intl*	2099.HK	Not Rated		177.40	NA	8,772	11.3	11.0	8.5	8.6	3.5	2.8	31.3	26.1	NA	NA
H-share avg.							**8.9**	**8.3**	**5.9**	**5.5**	**2.4**	**2.0**	**29.8**	**25.4**	**3.5**	**3.9**
Southern Copper	SCCO.N	Sell	Copper	207.42	180.00	170,437	26.1	28.0	15.1	16.0	12.2	13.8	52.4	46.3	2.0	4.5
Freeport	FCX.N	Buy	Copper	72.42	88.00	104,574	19.9	15.3	11.2	8.7	4.5	3.6	25.0	26.2	0.8	0.8
Antofagasta	ANTO.L	Neutral	Copper	3,588	4,200	46,892	19.4	18.6	10.1	9.3	4.1	3.8	23.1	21.1	2.6	2.7
First Quantum	FM.TO	Buy	Copper	39.00	60.00	22,828	NA	14.0	12.0	6.6	1.9	1.7	4.6	12.9	-	-
Sandfire	SFR.AX	Neutral	Copper	22.38	23.30	7,069	14.3	13.5	5.7	5.5	2.3	2.9	17.8	22.4	2.2	2.2
Ivanhoe	IVN.TO	Buy	Copper	11.20	16.00	11,222	31.1	15.2	13.0	8.3	1.8	1.6	6.1	11.2	-	-
Aurubis	NAFG.DE	Neutral	Copper recycle	158.40	170.00	7,744	16.9	16.2	8.9	8.2	1.1	1.1	7.4	6.9	-	1.8
BHP	BHPB.L	Neutral	Diversified	3,193	3,250	214,857	16.5	16.9	8.4	8.3	4.2	3.9	26.8	24.0	5.7	4.2
Rio Tinto	RIO.L	Neutral	Diversified	6,997	7,300	150,635	11.7	12.0	6.8	6.8	2.2	2.0	19.9	17.5	5.2	5.0
Glencore	GLEN.L	Buy	Diversified	582.90	690	90,447	11.2	11.1	4.8	4.5	2.1	1.8	19.8	17.4	3.2	3.5
Anglo American	AAL.L	Buy	Diversified	4,120	4,800	58,937	28.9	16.6	10.5	7.9	3.3	2.5	11.4	20.5	8.9	2.4
Global copper avg.							**19.1**	**16.1**	**9.7**	**8.2**	**3.6**	**3.5**	**19.5**	**20.6**	**2.8**	**2.5**

Source: UBS, Wind, Thompson Reuters; Data as of Oct 13 2026

UBS HOLT analysis

We use UBS HOLT to examine what investors are implicitly assuming about future corporate returns. UBS HOLT provides a consistent, accounting-adjusted framework for comparing corporate profitability and valuation globally. CFROI approximates the real economic return generated by a company's operating asset base.

What is UBS HOLT?

UBS HOLT is a data and analytics platform that helps investors make better decisions by using an objective framework for comparing and valuing companies. HOLT provides investors with a unique and consistent approach to benchmarking the corporate profitability of 20,000 companies around the world. The HOLT methodology eliminates distortions in global accounting statements, enabling clients to better understand the real economic drivers of performance.

What is CFROI?

HOLT uses the Cash Flow Return on Investment (CFROI) as an approximation of the economic returns earned by a firm on the portfolio of projects that constitute its operating assets. It is an estimate of the average real internal rate of a firm, recognizing the finite economic life of depreciating assets and the residual value of non-depreciating assets.

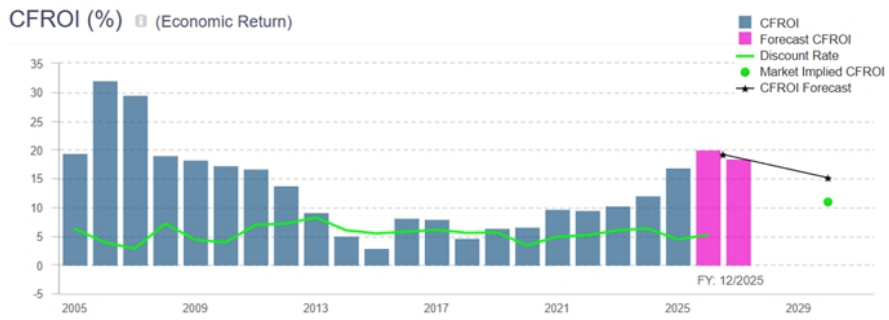
Market-implied CFROI still points to cyclical mean reversion

The market is pricing in a much faster normalisation in returns than our forecasts suggest. For HK/China copper miners, forecast CFROI rises to near two-decade highs over the next two years, supported by higher commodity prices, production growth and stronger profitability. For Zijin Mining, CMOC and Jiangxi Copper, market-implied CFROIs, shown as green dots, remain close to their historical mid-cycle levels, represented by the five-year medians. This suggests investors expect elevated near-term returns to fade relatively quickly. In our view, current valuations overstate the risk of mean reversion and give limited credit to stronger-for-longer copper prices and earnings, leaving room for a re-rating if returns remain elevated.

Zijin shows the clearest expectations gap

Zijin generates the highest forecast CFROI among the HK/China peers in the analysis, supported by production growth, competitive costs and a broad earnings base. Yet its market-implied CFROI remains toward the lower end of its recent range and materially below the near-term forecast. This suggests that the market is giving limited credit to the sustainability of Zijin's returns.The contrast with Southern Copper is notable. Southern Copper's market-implied CFROI has risen materially and appears to incorporate a meaningful improvement in profitability, in some cases exceeding near-term forecast levels. Zijin, by contrast, is priced for a faster return toward historical profitability despite offering stronger volume growth.

Figure 49: Zijin's historical CFROI

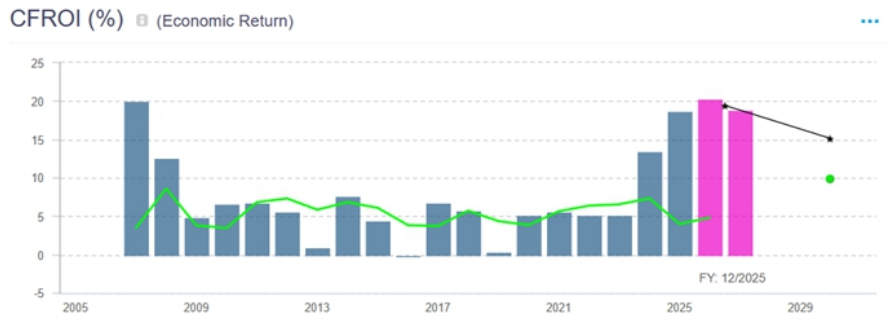


Source: UBS HOLT

CMOC also embeds conservative long-term expectations

CMOC's forecast CFROI benefits from volume growth and high-grade assets, but its market-implied return remains below the forecast trajectory. This indicates that investors continue to discount the durability of earnings, potentially reflecting concerns around jurisdiction, taxation, cost inflation and cobalt-price volatility. These risks justify some discount, but continued volume delivery and disciplined cost control could narrow the gap between forecast and market-implied returns. CMOC therefore has scope to re-rate through both earnings delivery and a higher valuation of those earnings.

Figure 50: CMOC's historical CFROI

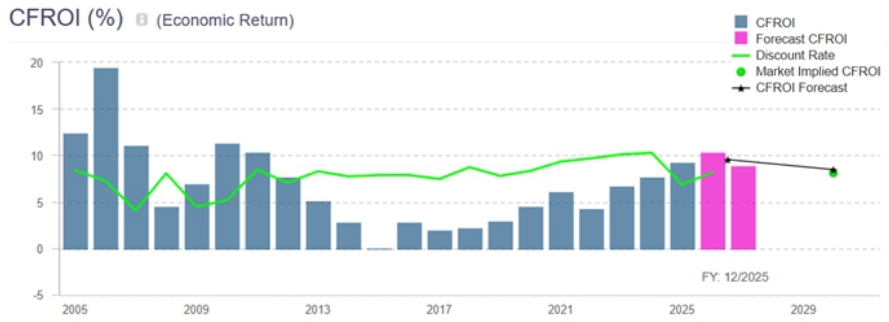


Source: UBS HOLT

Jiangxi Copper's low expectations reflect its mixed business model

Jiangxi Copper's CFROI profile is influenced by its combination of mining, smelting and downstream operations. Its lower organic mining growth and exposure to depressed TC/RCs help explain why the market applies a more conservative return assumption. However, the current framework may give insufficient credit to potential earnings from First Quantum, SolGold, tungsten and copper foil.For Jiangxi Copper, re-rating may depend less on broad multiple convergence and more on the market recognising these separate sources of asset value and potential earnings upside.

Figure 51: Jiangxi Copper's historical CFROI



Source: UBS HOLT

The peer-level UBS HOLT comparison reinforces the valuation gap shown by conventional multiples. Relative to their near-term CFROI forecasts, Zijin Mining carries the lowest market-implied expectations in the peer group, followed by CMOC and Jiangxi Copper. Zijin's market-implied CFROI remains near the lower end of its five-year range and well below its forecast, indicating that the market gives limited credit to the sustainability of its stronger returns.

By contrast, overseas peers such as Southern Copper have already experienced a material re-rating. Its market-implied CFROI has risen over the past 12 months to above its near-term forecast, suggesting that the current valuation already reflects confidence in stronger-for-longer copper profitability. The contrast is particularly striking for Zijin, which offers stronger production growth and comparable operating quality, but carries substantially lower market-implied expectations.

We therefore see the strongest re-rating case in Zijin, where the gap between operating quality, forecast returns and market expectations is the widest. CMOC also embeds conservative expectations despite its high-grade resources and sustained volume growth, while Jiangxi Copper offers lower-multiple exposure to First Quantum, SolGold and other asset and earnings catalysts. Together, the three names offer differentiated exposure to the same central thesis: copper is becoming more structural, while HK/ China copper miners remain priced for cyclical mean reversion.

Figure 52: HK/China miners embed materially lower return expectations than overseas peers



Source: UBS HOLT

APAC Focus represents a curated selection of the best insights from Global Research APAC and showcases differentiated views and actionable ideas, based around unique evidence and key industry themes. It leverages the combined efforts of 130+ APAC analysts, and colleagues in other regions, to develop a broad and collaborative perspective. This research often leverages UBS proprietary assets, including UBS Evidence lab and Quant Answers, whilst utilizing new analytical approaches and identifying critical ESG themes.

To learn more about APAC Focus, visit UBS Neo.

Valuation Method and Risk Statement

Copper sector - Downside risks include: 1) worsened property construction; 2) less-than-expected solar/wind installation; 3) declining home appliance export order; 4) State Grid and Southern Grid capex missed. Upside risks include: 1) improved property construction, 2) better-than-expected solar/wind installation, 3) State Grid and Southern Grid capex beat, 4) home appliance shipments beat.
Zijin price target is based on a SOTP multiple. Zijin's company-specific risks include but are not limited to: overseas political risks; the prices of two commodities (copper and gold); faster-/slower-than-expected ramp ups of new mines; and FX risks.
CMOC price target on an EV/EBITDA multiple. Specific risks to CMOC include but are not limited to: overseas political risks, commodity price risks (especially for copper and cobalt), a faster/slower-than-expected ramp-up in new mines, and forex risk.
JCC price target is based on SOTP. We believe the key company-specific risk is the outcome of FM's interest in Cobre Panama, where the company has invested ~$10bn to date. We also see various outcomes for FM to manage liquidity risks while the mine is non-operational via cuts to capex, potential asset sales, etc.
The HOLT methodology does not assign ratings or a target price to a security. It is an analytical tool that involves use of a set of proprietary quantitative algorithms and warranted value calculations, collectively called the HOLT valuation model, that are consistently applied to all the companies included in its database. The HOLT valuation model is a discounted cash flow model. Third-party data (including consensus earnings estimates) are systematically translated into a number of default variables and incorporated into the algorithms available in the HOLT valuation model. The source financial statement, pricing, and earnings data provided by outside data vendors are subject to quality control and may also be adjusted to more closely measure the underlying economics of firm performance. These adjustments provide consistency when analyzing a single company across time, or analyzing multiple companies across industries or national borders.
The default scenario that is produced by the HOLT valuation model establishes a warranted price that represents the expected mean value for a security based upon empirically derived fade algorithms that forecast a firm's future return on capital and growth rates over an extended period of time. As the third-party data are updated, the warranted price updates automatically. A company's future achieved return on capital or growth rate may differ from HOLT default forecast. Additional information about the HOLT methodology is available upon request.
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HOLT transferred to UBS on October 1, 2024 – renamed as UBS HOLT. The UBS HOLT Lens website and materials will be transitioning to UBS branding over time, with expected full redesign to UBS in time.

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12-Month Rating	Definition	Coverage[1]	IB Services[2]
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	39%	20%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 September 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

KEY DEFINITIONS:Forecast Stock Return (FSR) is defined as expected percentage price appreciation plus gross dividend yield over the next 12 months. In some cases, this yield may be based on accrued dividends. **Market Return Assumption (MRA)** is defined as the one-year local market interest rate plus 5% (a proxy for, and not a forecast of, the equity risk premium). **Under Review (UR)** Stocks may be flagged as UR by the analyst, indicating that the stock's price target and/or rating are subject to possible change in the near term, usually in response to an event that may affect the investment case or valuation. **Equity Price Targets** have an investment horizon of 12 months.

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Company Disclosures

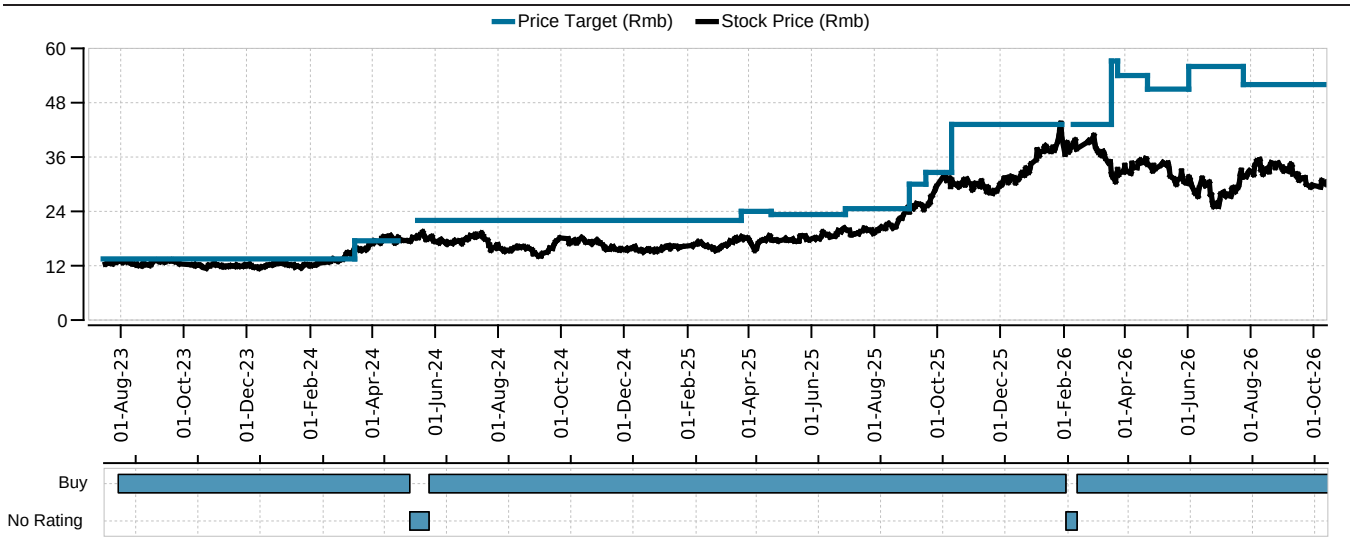
Company Name	Reuters	12-month rating	Price	Price date
CMOC[13,16,7,18]	603993.SS	Buy	Rmb16.64	14 Oct 2026
CMOC - H[13,16,7,18]	3993.HK	Buy	HK$15.23	14 Oct 2026
Jiangxi Copper[16,28,18]	0358.HK	Buy	HK$32.58	14 Oct 2026
Zijin Mining Group - A[2,13,4,16,6,28,18]	601899.SS	Buy	Rmb29.98	14 Oct 2026
Zijin Mining Group - H[2,13,4,16,6,28,18]	2899.HK	Buy	HK$32.06	14 Oct 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.

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Zijin Mining Group - A (Rmb)

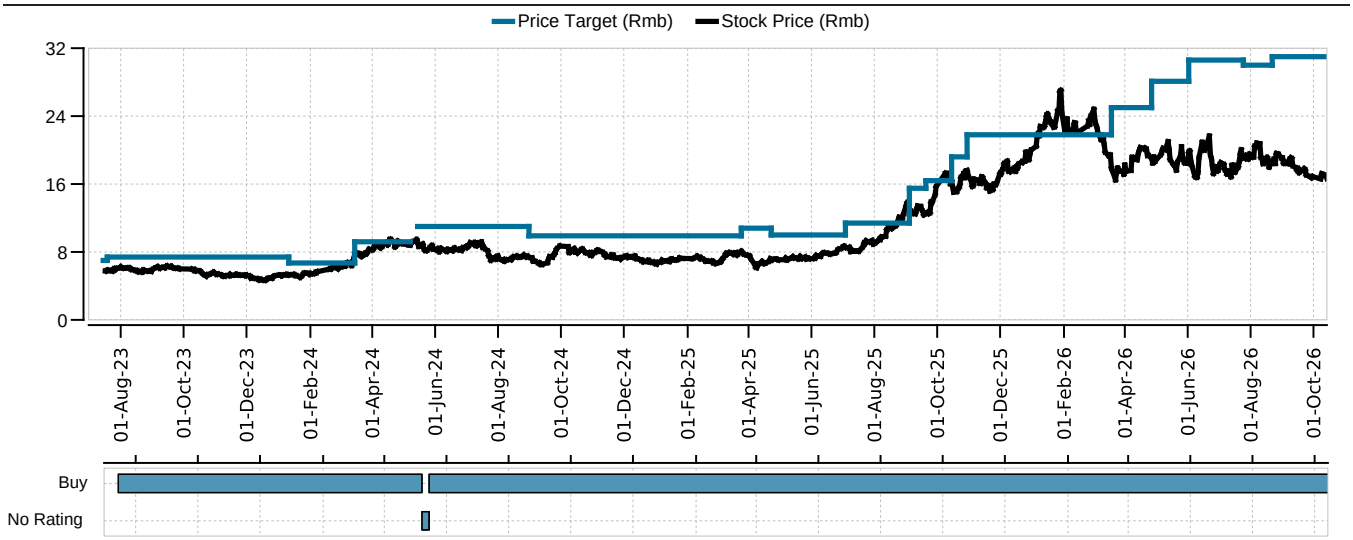


Date	Stock Price (Rmb)	Price Target (Rmb)	Rating
2023-07-14	12.58	13.50	Buy
2024-03-14	15.39	17.50	Buy
2024-04-25	17.40	-	No Rating
2024-05-14	18.49	22.00	Buy
2025-03-24	18.05	24.00	Buy

Date	Stock Price (Rmb)	Price Target (Rmb)	Rating
2025-04-22	18.50	23.30	Buy
2025-07-03	20.29	24.60	Buy
2025-09-03	24.94	30.00	Buy
2025-09-19	25.02	32.60	Buy
2025-10-14	29.50	43.20	Buy
2026-01-29	43.45	-	No Rating
2026-02-09	38.63	43.20	Buy
2026-03-18	34.92	57.20	Buy
2026-03-24	32.20	54.00	Buy
2026-04-22	35.53	51.00	Buy
2026-06-01	30.19	56.00	Buy
2026-07-24	31.51	52.00	Buy

Source: UBS Global Research; LSEG Eikon as of 14-Oct-2026. All prices as of local market close. Ratings as of date shown.

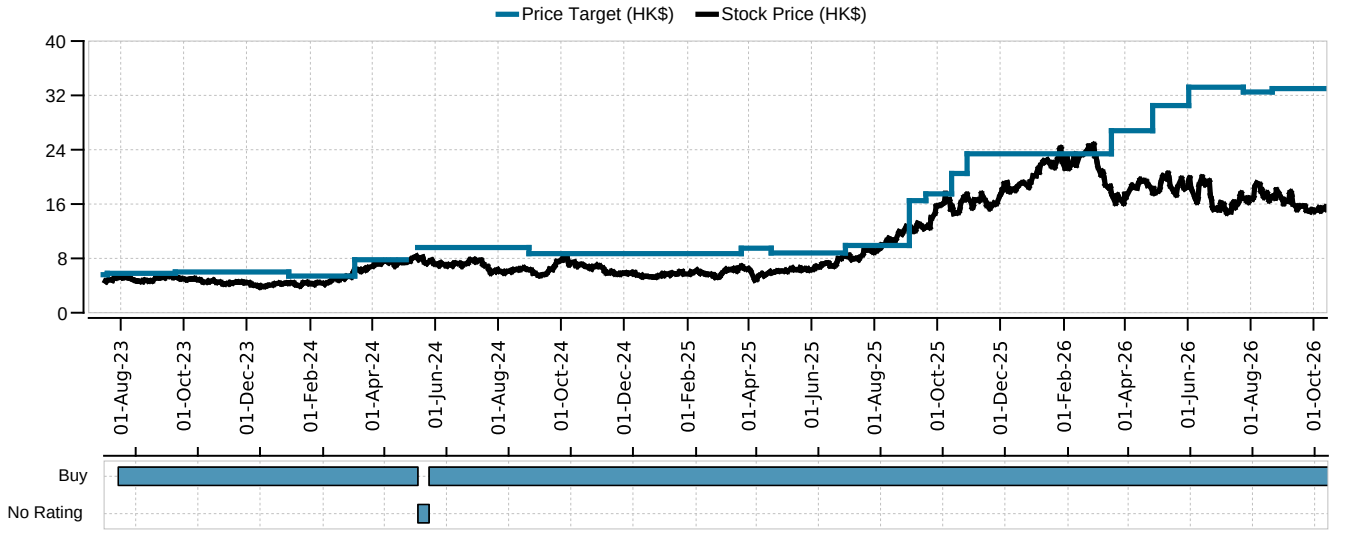
CMOC (Rmb)



Date	Stock Price (Rmb)	Price Target (Rmb)	Rating
2023-07-14	5.84	7.00	Buy
2023-07-18	5.75	7.40	Buy
2024-01-10	5.29	6.70	Buy
2024-03-14	7.19	9.20	Buy
2024-05-07	8.83	-	No Rating
2024-05-14	9.46	11.00	Buy
2024-08-30	7.49	9.90	Buy
2025-03-24	8.01	10.80	Buy
2025-04-22	7.20	10.00	Buy
2025-07-03	8.66	11.40	Buy
2025-09-03	13.65	15.50	Buy
2025-09-19	12.51	16.40	Buy
2025-10-14	16.01	19.20	Buy
2025-10-29	17.55	21.80	Buy
2026-03-18	19.41	25.00	Buy
2026-04-26	19.37	28.10	Buy
2026-06-01	18.42	30.60	Buy
2026-07-24	19.04	30.00	Buy
2026-08-21	18.62	31.00	Buy

Source: UBS Global Research; LSEG Eikon as of 14-Oct-2026. All prices as of local market close. Ratings as of date shown.

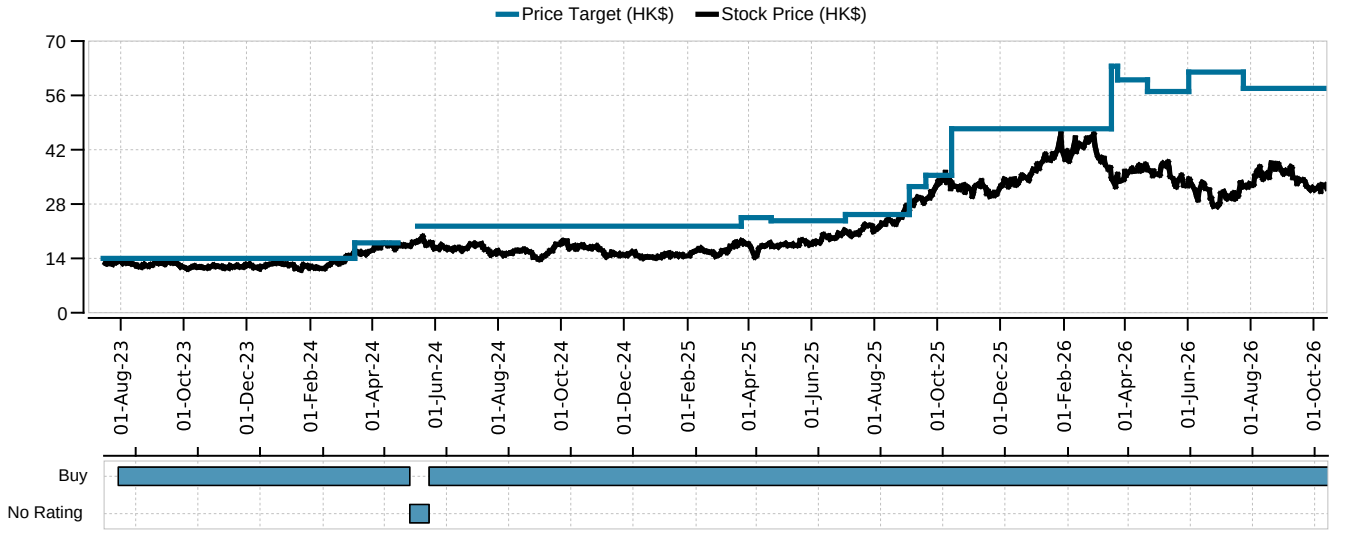
CMOC - H (HK$)



Date	Stock Price (HK$)	Price Target (HK$)	Rating
2023-07-14	4.66	5.60	Buy
2023-07-18	4.58	5.80	Buy
2023-09-22	5.24	6.00	Buy
2024-01-10	4.38	5.40	Buy
2024-03-14	6.11	7.80	Buy
2024-05-03	7.40	-	No Rating
2024-05-14	8.34	9.60	Buy
2024-08-30	6.36	8.70	Buy
2025-03-24	6.72	9.50	Buy
2025-04-22	6.06	8.80	Buy
2025-07-03	8.43	9.90	Buy
2025-09-03	12.69	16.50	Buy
2025-09-19	12.49	17.50	Buy
2025-10-14	15.25	20.50	Buy
2025-10-29	16.65	23.40	Buy
2026-03-18	18.60	26.80	Buy
2026-04-27	18.25	30.50	Buy
2026-06-01	18.50	33.20	Buy
2026-07-24	16.82	32.50	Buy
2026-08-21	17.00	33.00	Buy

Source: UBS Global Research; LSEG Eikon as of 14-Oct-2026. All prices as of local market close. Ratings as of date shown.

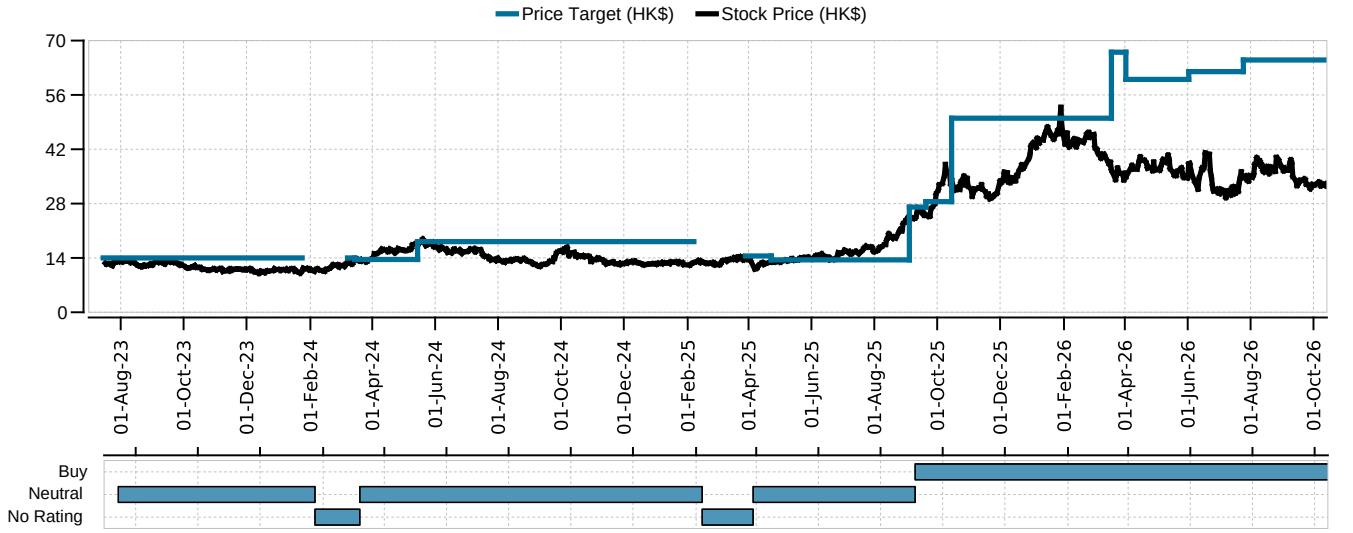
Zijin Mining Group - H (HK$)



Date	Stock Price (HK$)	Price Target (HK$)	Rating
2023-07-14	12.80	14.00	Buy
2024-03-14	15.46	18.00	Buy
2024-04-25	17.06	-	No Rating
2024-05-14	18.38	22.30	Buy
2025-03-24	18.14	24.50	Buy
2025-04-22	18.02	23.70	Buy
2025-07-03	21.25	25.30	Buy
2025-09-03	27.60	32.50	Buy
2025-09-19	29.08	35.40	Buy
2025-10-14	32.00	47.40	Buy
2026-03-18	37.34	63.50	Buy
2026-03-24	34.86	60.00	Buy
2026-04-22	37.84	57.00	Buy
2026-06-01	32.76	62.00	Buy
2026-07-24	32.38	57.80	Buy

Source: UBS Global Research; LSEG Eikon as of 14-Oct-2026. All prices as of local market close. Ratings as of date shown.

Jiangxi Copper (HK$)



Date	Stock Price (HK$)	Price Target (HK$)	Rating
2023-07-14	12.68	14.00	Neutral

Date	Stock Price (HK$)	Price Target (HK$)	Rating
2024-01-23	10.30	-	No Rating
2024-03-07	12.58	14.00	Neutral
2024-03-14	13.32	13.60	Neutral
2024-05-14	17.90	18.20	Neutral
2025-02-06	12.70	-	No Rating
2025-03-28	14.24	14.50	Neutral
2025-04-22	12.90	13.50	Neutral
2025-09-03	24.44	27.10	Buy
2025-09-19	25.38	28.50	Buy
2025-10-14	33.10	50.00	Buy
2026-03-18	38.66	67.00	Buy
2026-04-01	35.84	60.00	Buy
2026-06-01	34.62	62.00	Buy
2026-07-24	34.62	65.00	Buy

Source: UBS Global Research; LSEG Eikon as of 14-Oct-2026. All prices as of local market close. Ratings as of date shown.

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